

```
In [2]: 1 import array
        2 import collections
        3 import copy
        4 import decimal
        5 import functools
        6 import importlib
        7 import logging
        8 import math
        9 import pprint
       10 import random
       11 import re
       12 import statistics
       13 import string
       14 import sys
       15 import time
       16 import timeit
       17
       18 import pandas
```

```
In [2]: 1 num1 = 10_00_000
        2 print(num1)
```

```
1000000
```

```
In [3]: 1 print(format(0.1, ".25f"))
```

```
0.1000000000000000055511151
```

```
In [4]: 1 print(format(0.125, ".25f"))
```

```
0.12500000000000000000000000
```

```
In [5]: 1 print(0.1 + 0.1 + 0.1 == 0.3)
```

```
False
```

```
In [6]: 1 print(format(0.3, ".25f"))
```

```
0.2999999999999999888977698
```

```
In [7]: 1 print(format(0.2, ".25f"))
```

```
0.200000000000000011022302
```

```
In [8]: 1 print(abs((0.1 + 0.1 + 0.1) - 0.3) < 0.001)
```

```
True
```

```
In [9]: 1 print((0.125).as_integer_ratio())
```

```
(1, 8)
```

```
In [10]: 1 print((10).          (25))
```

35

```
In [11]: 1 a = float(10)
         2 print(a)
```

10.0

```
In [12]: 1 c = (-4) ** 0.5
         2 print(c)
```

(1.2246467991473532e-16+2j)

```
In [13]: 1 print(type(c))
```

<class 'complex'>

```
In [14]: 1 print(c.real)
```

1.2246467991473532e-16

```
In [15]: 1 print(c.imag)
```

2.0

```
In [16]: 1 c = 10 + 2j
         2 print(type(c))
```

<class 'complex'>

```
In [17]: 1 print(c)
```

(10+2j)

```
In [18]: 1 try:
         2     print("2" < 2)
         3 except TypeError as ex:
         4     print(type(ex).    )
```

TypeError

```
In [19]: 1 print("2" == 2)
```

False

```
In [20]: 1 a = 0
         2 b = 1
         3 c = 500
         4 d = 0
         5 e = 1
         6 f = 500
         7 print(id(a), id(b), id(c), id(d), id(e), id(f))
```

11370000 11370032 125017714239248 11370000 11370032 125017714237744

```
In [21]: 1 print(0.1 * 3 - 0.3 < abs(1e-6))
```

True

```
In [22]: 1 a = 257
        2 print(a is 257)
```

False

<>:2: SyntaxWarning: "is" with 'int' literal. Did you mean "=="?

<>:2: SyntaxWarning: "is" with 'int' literal. Did you mean "=="?

/tmp/ipykernel_1151058/1958631963.py:2: SyntaxWarning: "is" with 'int' literal. Did you mean "=="?
print(a is 257)

```
In [23]: 1 a = 10
        2 b = 0
        3 print(b != 0 and a / b)
```

False

```
In [24]: 1 a = 1e-6
        2 b = 2e-6
        3 tol = 1e-7
        4 print(abs(a - b) <= tol)
```

False

```
In [25]: 1 elapsed = 7835
        2 hours = elapsed // 3600
        3 minutes = (elapsed - (hours * 3600)) // 60
        4 seconds = elapsed - ((hours * 3600) + (minutes * 60))
        5 print(hours, minutes, seconds)
```

2 10 35

```
In [26]: 1 elapsed = 7835
        2 hours = elapsed // (60 * 60)
        3 minutes = (elapsed % (60 * 60)) // 60
        4 seconds = (elapsed % (60 * 60)) % 60
        5 print(hours, minutes, seconds)
```

2 10 35

```
In [27]: 1 a = None
        2 if a == None:
        3     a = "N/A"
        4 print(a)
```

N/A

```
In [28]: 1 a = None
        2 a = "N/A" if a is None else a
        3 print(a)
```

N/A

```
In [29]: 1 score = 700
        2 rating = "Invalid"
        3 if score < 0:
        4     rating = "Negative"
```

```

5 elif score < 580:
6     rating = "Poor"
7 elif score < 670:
8     rating = "Fair"
9 elif score < 740:
10    rating = "Good"
11 elif score < 800:
12    rating = "Very Good"
13 elif score < 850:
14    rating = "Excellent"
15 print(rating)

```

Good

```

In [30]: 1 elapsed = 1000000
          2 magnitude = "Invalid"
          3 if elapsed < 60:
          4     magnitude = "seconds"
          5 elif elapsed < (60 * 60):
          6     magnitude = "minutes"
          7 elif elapsed < (60 * 60 * 24):
          8     magnitude = "hours"
          9 elif elapsed < (60 * 60 * 24 * 7):
         10     magnitude = "days"
         11 elif elapsed > (60 * 60 * 24 * 7):
         12     magnitude = "weeks"
         13 print(magnitude)

```

weeks

```

In [31]: 1 t = [1, 2], 3, 4
          2 l = list(t)
          3 print(l)

```

[[1, 2], 3, 4]

```

In [32]: 1 l[0][1] = 20
          2 print(l)

```

[[1, 20], 3, 4]

```

In [33]: 1 print(t)

```

([1, 20], 3, 4)

```

In [34]: 1 for i in t:
          2     print(id(i))

```

125017713390016
11370096
11370128

```

In [35]: 1 for i in l:
          2     print(id(i))

```

```
125017713390016
11370096
11370128
```

```
In [36]: 1 for i, j in zip(l, t):
          2     print(i is j)
```

```
True
True
True
```

```
In [37]: 1 s = "ABCDEFGHJIJ"
          2 print(s[3:6:-1])
```

```
In [38]: 1 print(s[3:5 + 1])
```

```
DEF
```

```
In [39]: 1 m1 = [[0, 0, 0], [0, 0, 0], [0, 0, 0]]
          2 m2 = m1[0:2]
          3 print(m1[0] is m2[1])
```

```
False
```

```
In [40]: 1 m1[0] = "Python"
          2 print(m1)
```

```
['Python', [0, 0, 0], [0, 0, 0]]
```

```
In [41]: 1 print(m2)
```

```
[[0, 0, 0], [0, 0, 0]]
```

```
In [42]: 1 m1[1][0] = 100
```

```
In [43]: 1 print(m1)
```

```
['Python', [100, 0, 0], [0, 0, 0]]
```

```
In [44]: 1 print(m2)
```

```
[[0, 0, 0], [100, 0, 0]]
```

```
In [45]: 1 print("abcdef"[-1:-4])
```

```
In [46]: 1 print("abcdef"[1:4:-1])
```

```
In [47]: 1 print("racecar" == "racecar"[::-1])
```

```
True
```

```
In [48]: 1 list1 = [0, 1, 2, 3, 4]
          2 list1[0:3] = "abc"
```

```
3 print(list1)
```

```
['a', 'b', 'c', 3, 4]
```

```
In [49]: 1 list1[0:3] = ["a", "b", "c"]  
2 print(list1)
```

```
['a', 'b', 'c', 3, 4]
```

```
In [50]: 1 list1[0:3] = ["a", "b"]  
2 print(list1)
```

```
['a', 'b', 3, 4]
```

```
In [51]: 1 del list1[2]  
2 print(list1)
```

```
['a', 'b', 4]
```

```
In [52]: 1 del list1[0:2]  
2 print(list1)
```

```
[4]
```

```
In [53]: 1 list1.extend("ABC")  
2 print(list1)
```

```
[4, 'A', 'B', 'C']
```

```
In [54]: 1 list1.insert(0, "Y0!")  
2 print(list1)
```

```
['Y0!', 4, 'A', 'B', 'C']
```

```
In [55]: 1 list1[:-3:-1] = "cb"  
2 print(list1)
```

```
['Y0!', 4, 'A', 'b', 'c']
```

```
In [56]: 1 l = []  
2 print(timeit.timeit("l.append(1)", globals = globals(), number = 100_000))
```

```
0.0027827439989778213
```

```
In [57]: 1 print(len(l))
```

```
100000
```

```
In [58]: 1 list2 = []  
2 print(timeit.timeit("list2.insert(0, 1)", globals = globals(), number =  
100_000))
```

```
2.075601356002153
```

```
In [59]: 1 print(len(list2))
```

```
100000
```

```
In [60]: 1 print(1.249788 / 0.0018043)
```

```
692.6719503408524
```

```
In [61]: 1 list2 = []  
2 print(timeit.timeit("list2.insert(len(list2), 1)", globals = globals(), number  
= 100_000))
```

```
0.005791994997707661
```

```
In [62]: 1 print(0.0057634 / 0.0018043)
```

```
3.194258161059691
```

```
In [63]: 1 l1 = [[0, 1], 2, "3"]  
2 l2 = l1[:]  
3 for i in l1:  
4     print(id(i))
```

```
125017713495424  
11370064  
11434512
```

```
In [64]: 1 for i in l2:  
2     print(id(i))
```

```
125017713495424  
11370064  
11434512
```

```
In [65]: 1 l2[0][0] = "0"  
2 for i in l1:  
3     print(id(i))
```

```
125017713495424  
11370064  
11434512
```

```
In [66]: 1 for i in l2:  
2     print(id(i))
```

```
125017713495424  
11370064  
11434512
```

```
In [67]: 1 for i in l1:  
2     print(i)
```

```
['0', 1]  
2  
3
```

```
In [68]: 1 for i in l2:  
2     print(i)
```

```
['0', 1]  
2  
3
```

```
In [69]: 1 l2[1] = "22"  
2 for i in l1:  
3     print(id(i))
```

```
125017713495424  
11370064  
11434512
```

```
In [70]: 1 for i in l2:  
2         print(id(i))
```

```
125017713495424  
125019187127488  
11434512
```

```
In [71]: 1 for i in l1:  
2         print(i)
```

```
['0', 1]  
2  
3
```

```
In [72]: 1 for i in l2:  
2         print(i)
```

```
['0', 1]  
22  
3
```

```
In [73]: 1 l1 = [[0, 1], 2, "3"]  
2 l2 = l1[:]  
3 l2[0] = ["0", 1]  
4 for i in l1:  
5     print(id(i))
```

```
125017713508416  
11370064  
11434512
```

```
In [74]: 1 for i in l2:  
2         print(id(i))
```

```
125017713510656  
11370064  
11434512
```

```
In [75]: 1 for i in l1:  
2         print(i)
```

```
[0, 1]  
2  
3
```

```
In [76]: 1 for i in l2:
         2     print(i)
```

```
['0', 1]
2
3
```

```
In [77]: 1 l1 = [[0, 1], 2, "3"]
         2 l2 = l1[:]
         3 print(id(l1), id(l2))
```

```
125017713510720 125017713510592
```

```
In [78]: 1 for i, j in zip(l1, l2):
         2     print(id(i), id(j))
```

```
125017713510528 125017713510528
11370064 11370064
11434512 11434512
```

```
In [79]: 1 for i, j in zip(l1[0], l2[0]):
         2     print(id(i), id(j))
```

```
11370000 11370000
11370032 11370032
```

```
In [80]: 1 l1 = [[0, 1], 2, "3"]
         2 l2 = l1.copy()
         3 print(id(l1), id(l2))
```

```
125017713508928 125017713510016
```

```
In [81]: 1 for i, j in zip(l1, l2):
         2     print(id(i), id(j))
```

```
125018848286208 125018848286208
11370064 11370064
11434512 11434512
```

```
In [82]: 1 for i, j in zip(l1[0], l2[0]):
         2     print(id(i), id(j))
```

```
11370000 11370000
11370032 11370032
```

```
In [83]: 1 l1 = [[0, 1], 2, "3"]
         2 l2 = copy.deepcopy(l1)
         3 print(id(l1), id(l2))
```

```
125017713244416 125017713512384
```

```
In [84]: 1 for i, j in zip(l1, l2):
         2     print(id(i), id(j))
```

```
125017713514496 125017713386752
11370064 11370064
11434512 11434512
```

```
In [85]: 1 for i, j in zip(l1[0], l2[0]):
          2     print(id(i), id(j))
```

```
11370000 11370000
11370032 11370032
```

```
In [86]: 1 t1 = (0, "1", "23")
          2 t2 = t1[:]
          3 print(t1 is t2)
```

```
True
```

```
In [87]: 1 a, b, c, d = "abcd"
          2 print(a, b, c, d)
```

```
a b c d
```

```
In [88]: 1 s = "FfEeDdCcBbAa"
          2 lowerLetters = s[::-2]
          3 upperLetters = s[-2::-2]
          4 print(lowerLetters)
```

```
abcdef
```

```
In [89]: 1 print(upperLetters)
```

```
ABCDEF
```

```
In [90]: 1 t1 = 1, 2, 3, 4, 5, 6
          2 t2 = 7, 8, 9, 10
          3 t3 = 11, 12, 13, 14, 15, 16, 17
          4 t = t1 + t2 + t3
          5 l = list(t)
          6 l[::2] = [0] * len(l[::2])
          7 print(tuple(l))
```

```
(0, 2, 0, 4, 0, 6, 0, 8, 0, 10, 0, 12, 0, 14, 0, 16, 0)
```

```
In [91]: 1 m = [[0, 0, 0], [0, 0, 0], [0, 0, 0]]
          2 m[0][0] = 1
          3 m[1][1] = 1
          4 m[2][2] = 1
          5 print(m)
```

```
[[1, 0, 0], [0, 1, 0], [0, 0, 1]]
```

```
In [92]: 1 m = [[0, 0, 0], [0, 0, 0], [0, 0, 0]]
          2 m1 = copy.deepcopy(m)
          3 m1[0][0] = 1
          4 m1[1][1] = 1
```

```
5 m1[2][2] = 1
6 print(m1)
```

```
[[1, 0, 0], [0, 1, 0], [0, 0, 1]]
```

```
In [93]: print(m)
1
2 data = [(100, "USD", "EUR", 0.83), (100, "USD", "CAD", 1.27), (100, "CAD",
3 "EUR", 0.65)]
4 USDAmountInEUR = data[0][0] * data[0][3]
5 print(str(data[0][0]) + " " + data[0][1] + " = " + str(USDAmountInEUR) + " " +
str(data[0][2]))
```

```
[[0, 0, 0], [0, 0, 0], [0, 0, 0]]
100 USD = 83.0 EUR
```

```
In [94]: 1 USDAmountInCAD = data[1][0] * data[1][3]
2 print(str(data[1][0]) + " " + data[1][1] + " = " + str(USDAmountInCAD) + " " +
str(data[1][2]))
```

```
100 USD = 127.0 CAD
```

```
In [95]: 1 CADAmountInEUR = data[2][0] * data[2][3]
2 print(str(data[2][0]) + " " + data[2][1] + " = " + str(CADAmountInEUR) + " " +
str(data[2][2]))
```

```
100 CAD = 65.0 EUR
```

```
In [96]: 1 print(ord("A"))
```

```
65
```

```
In [97]: 1 print("\N{SNAKE}")
```

```
🐍
```

```
In [98]: 1 print(ord("\N{SNAKE}"))
```

```
128013
```

```
In [99]: 1 print(hex(ord("\N{SNAKE}")))
```

```
0x1f40d
```

```
In [100]: 1 print("\u03b1")
```

```
α
```

```
In [101]: 1 print("\U0001f40d")
```

```
🐍
```

```
In [102]: 1 print(int("1f40d", 16))
```

```
128013
```

```
In [103]: 1 print("\N[SNAKE] is a snake")
```

```
🐍 is a snake
```

```
In [104]: 1 print(int("7ff", 16))
```

```
2047
```

```
In [105]: 1 print(int("ffff", 16))
```

```
65535
```

```
In [106]: 1 print(int("10ffff", 16))
```

```
1114111
```

```
In [107]: 1 print(bin(2047))
```

```
0b1111111111
```

```
In [108]: 1 print(len(bin(2047)))
```

```
13
```

```
In [109]: 1 print(bin(65535))
```

```
0b1111111111111111
```

```
In [110]: 1 print(len(bin(65535)))
```

```
18
```

```
In [111]: 1 print(bin(1114111))
```

```
0b10000111111111111111
```

```
In [112]: 1 print(len(bin(1114111)))
```

```
23
```

```
In [113]: 1 print(bin(128013))
```

```
0b11111010000001101
```

```
In [114]: 1 print("\n{snake}")
```

```
{snake}
```

```
In [115]: 1 a = True  
2 b = True  
3 print(a is b)
```

```
True
```

```
In [116]: 1 a = False
          2 b = False
          3 print(a is b)
```

True

```
In [117]: 1 a = True
          2 print(id(a))
```

11120192

```
In [118]: 1 a = False
          2 print(id(a))
```

11045056

```
In [119]: 1 b = False
          2 print(id(b))
```

11045056

```
In [120]: 1 ?
```

Docstring:

Return the lowest index in S where substring sub is found, such that sub is contained within S[start:end].

Optional arguments start and end are interpreted as in slice notation.
Return -1 on failure.

Type: method_descriptor

```
In [121]: 1 print(f"1 + 1 = {1 + 1}")
```

1 + 1 = 2

```
In [122]: 1 bid = 1.5760
          2 ask = 1.5763
          3 print("bid: {}, ask: {}, spread: {}".format(bid, ask, ask - bid))
```

bid: 1.576, ask: 1.5763, spread: 0.000299999999999996696

```
In [123]: 1 print("bid: {:.4f}, ask: {:.4f}, spread: {:.4f}".format(bid, ask, ask - bid))
```

bid: 1.5760, ask: 1.5763, spread: 0.0003

```
In [124]: 1 print(f"bid: {bid:.4f}, ask: {ask:.4f}, spread: {ask - bid:.4f}")
```

bid: 1.5760, ask: 1.5763, spread: 0.0003

```
In [125]: 1 print(f"bid: {bid:.4f}, ask: {ask:.4f}, spread: {(ask - bid):.4f}")
```

bid: 1.5760, ask: 1.5763, spread: 0.0003

```
In [126]: 1 string1 = "β, β, #"
          2 string1 = "Π, ú, θ, ω, ν"
```

```
3 chars = string1.split(", ")
4 print(chars)
```

```
['Π', 'ó', 'θ', 'ω', 'ν']
```

```
In [127]: 1 list1 = []
          2 for c in chars:
          3     list1.append(hex(ord(c)))
          4 print(list1)
```

```
['0x3a0', '0x3cd', '0x3b8', '0x3c9', '0x3bd']
```

```
In [128]: 1 list1 = []
          2 for c in chars:
          3     list1.append(c.upper())
          4 print(list1)
```

```
['Π', 'Υ', 'Θ', 'Ω', 'Ν']
```

```
In [129]: 1 list1 = []
          2 for c in chars:
          3     list1.append(c.lower())
          4 print(list1)
```

```
['π', 'ó', 'θ', 'ω', 'ν']
```

```
In [130]: 1 a = 42
          2 print(f"The number {a} is not even") if a % 2 else print(f"The number {a} is even")
```

```
The number 42 is even
```

```
In [131]: 1 a = 41
          2 print(f"The number {a} is not even") if a % 2 else print(f"The number {a} is even")
```

```
The number 41 is not even
```

```
In [132]: 1 a = 42
          2 print("The number {} is not even".format(a)) if a % 2 else print("The number {} is even".format(a))
```

```
The number 42 is even
```

```
In [133]: 1 a = 41
          2 print("The number {} is not even".format(a)) if a % 2 else print("The number {} is even".format(a))
```

```
The number 41 is not even
```

```
In [134]: 1 a = 0.3
          2 b = 0.1
          3 print(f"{a} / {b} = {a / b:.4f}")
```

```
0.3 / 0.1 = 3.0000
```

```
In [135]: 1 a = 42
          2 qualifier = "is not" if a % 2 else "is"
          3 print(f"The number {a} {qualifier} even")
```

The number 42 is even

```
In [136]: 1 a = 421
          2 qualifier = "is not" if a % 2 else "is"
          3 print(f"The number {a} {qualifier} even")
```

The number 421 is not even

```
In [137]: 1 print(range(10))
```

range(0, 10)

```
In [138]: 1 print(type(range(10)))
```

<class 'range'>

```
In [139]: 1 for i in range(3):
          2     print(i)
          3     print(i)
```

0
1
2
2

```
In [140]: 1 for i in "abc":
          2     pass
          3     print(i)
```

c

```
In [141]: 1 raggedArray = [[1, 2], [3, 4, 5, 6], [7, 8, 9], [10]]
          2 for r in range(len(raggedArray)):
          3     for c in range(len(raggedArray[r])):
          4         print(f"[{r}, {c}] = {raggedArray[r][c]}")
          5     print("-" * 5)
```

```

[0, 0] = 1
[0, 1] = 2
-----
[1, 0] = 3
[1, 1] = 4
[1, 2] = 5
[1, 3] = 6
-----
[2, 0] = 7
[2, 1] = 8
[2, 2] = 9
-----
[3, 0] = 10
-----

[0, 1] = 2
-----
[1, 0] = 3
[1, 1] = 4
[1, 2] = 5
[1, 3] = 6
-----
[2, 0] = 7
[2, 1] = 8
[2, 2] = 9
-----
[3, 0] = 10
-----

```

```

In [142]: 1 identityMatrix = []
          2 size = 10
          3 for r in range(size):
          4     row = []
          5     for c in range(size):
          6         if r == c:
          7             row.append(1)
          8         else:
          9             row.append(0)
         10     identityMatrix.append(row)
         11 identityMatrix

```

```

Out[142]: [[1, 0, 0, 0, 0, 0, 0, 0, 0, 0],
           [0, 1, 0, 0, 0, 0, 0, 0, 0, 0],
           [0, 0, 1, 0, 0, 0, 0, 0, 0, 0],
           [0, 0, 0, 1, 0, 0, 0, 0, 0, 0],
           [0, 0, 0, 0, 1, 0, 0, 0, 0, 0],
           [0, 0, 0, 0, 0, 1, 0, 0, 0, 0],
           [0, 0, 0, 0, 0, 0, 1, 0, 0, 0],
           [0, 0, 0, 0, 0, 0, 0, 1, 0, 0],
           [0, 0, 0, 0, 0, 0, 0, 0, 1, 0],
           [0, 0, 0, 0, 0, 0, 0, 0, 0, 1]]

```

```

In [143]: 1 raggedArray = [[1, 2], [3, 4, 5, 6], [7, 8, 9], [10]]
          2 for i, r in enumerate(raggedArray):
          3     for j, e in enumerate(raggedArray[i]): # e for element, cuz it's not a
          4         column technically
          5

```

```
print(f"[{i}, {j}] = {raggedArray[i][j]}")
print("-" * 8)
```

```
[0, 0] = 1
[0, 1] = 2
-----
[1, 0] = 3
[1, 1] = 4
[1, 2] = 5
[1, 3] = 6
-----
[2, 0] = 7
[2, 1] = 8
[2, 2] = 9
-----
[3, 0] = 10
-----
```

```
In [144]: 1 print(f"{0.1:.4f}")
```

```
0.1000
```

```
In [145]: 1 print("{:.4f}".format(0.1))
```

```
0.1000
```

```
In [146]: 1 total, count = (0,) * 2
2 data = [10.5, 11.2, 9.8, None, 11.5, None]
3 for e in data:
4     if e:
5         total += e
6         count += 1
7 average = total / count
8 print(average)
```

```
10.75
```

```
In [147]: 1 for i, e in enumerate(data):
2         if not e:
3             data[i] = average
4 print(data)
```

```
[10.5, 11.2, 9.8, 10.75, 11.5, 10.75]
```

```
In [148]: 1 data = [10.5, 11.2, 9.8, None, 11.5, None]
2 count = sum(1 for e in data if e)
3 total = sum(e for e in data if e)
4 average = total / count
5 print(sum, average)
```

```
<built-in function sum> 10.75
```

```
In [149]: 1 data = [10.5, 11.2, 9.8, None, 11.5, None]
2 average = statistics.fmean(e for e in data if e)
3 data = [average if e is None else e for e in data]
4 print(data)
```

```
[10.5, 11.2, 9.8, 10.75, 11.5, 10.75]
```

```
In [150]: 1 data = [10.5, 11.2, 9.8, None, 11.5, None]
          2 df = pandas.DataFrame(data)
          3 df.fillna(df.mean(), inplace = True)
          4 print(df[0].tolist())
```

```
[10.5, 11.2, 9.8, 10.75, 11.5, 10.75]
```

```
In [151]: 1 print(len(str((1, 2))))
```

```
6
```

```
In [152]: 1 print(str((1, 2)))
```

```
(1, 2)
```

```
In [153]: 1 m, n = 4, 5
          2 for i in range(1, m):
          3     for j in range(1, n):
          4         print(f"{i} x {j} = {i * j}")
          5     print("-" * 8)
```

```
1 x 1 = 1
1 x 2 = 2
1 x 3 = 3
1 x 4 = 4
-----
2 x 1 = 2
2 x 2 = 4
2 x 3 = 6
2 x 4 = 8
-----
3 x 1 = 3
3 x 2 = 6
3 x 3 = 9
3 x 4 = 12
-----
```

```
In [154]: 1 data = (["2021-01-01", 10, 20], ["2021-01-02", 20, 18], ["2021-01-03", -10,
          2 10], ["2021-01-04", 100, 102], ["2021-01-05", 20, 45])
          largest = 0
          3 for e in data:
          4     distance = abs(e[2] - e[1])
          5     e.append(distance)
          6     if distance > largest:
          7         dateOfLargest = e[0]
          8         largest = distance
          print(data)
          print(dateOfLargest)
```

9
1
0

```
(['2021-01-01', 10, 20, 10], ['2021-01-02', 20, 18, 2], ['2021-01-03', -10, 10, 20], ['2021-01-04', 100, 102, 2], ['2021-01-05', 20, 45, 25])  
2021-01-05
```

In [155]:

```
1 data = [[10, 20], [20, 30], [35, 50], [45, 60]]  
2 for i, r in enumerate(data):  
3     if i == 0:  
4         r.append("")  
5     else:  
6         if r[0] < data[i - 1][1]:  
7             r.append("down")  
8         elif r[0] > data[i - 1][1]:  
9             r.append("up")  
10        else:  
11            r.append("same")  
12 print(data)
```

```
[[10, 20, ''], [20, 30, 'same'], [35, 50, 'up'], [45, 60, 'down']]
```

In [156]:

```
1 tuple1 = ([0],)  
2 print(type(tuple1))  
3 print(tuple1)  
4 print(id(tuple1))  
5 print(id(tuple1[0]))  
6 tuple1[0][0] += 1  
7 print(tuple1)  
8 print(id(tuple1))  
9 print(id(tuple1[0]))
```

```
<class 'tuple'>  
([0],)  
125019117399616  
125017720340224  
([1],)  
125019117399616  
125017720340224
```

In [157]:

```
1 print(hash(True))  
2 a = True  
3 print(hash(a))  
4 b = True  
5 print(hash(b))  
6 a = b  
7 print(hash(a))
```

1
1
1
1

```
In [158]: 1 print(hash(257))
          2 a = 257
          3 print(hash(a))
          4 b = 257
          5 print(hash(b))
          6 a = b
          7 print(hash(a))
```

```
257
257
257
257
```

```
In [159]: 1 print(hash(2570))
          2 a = 2570
          3 print(hash(a))
          4 b = 2570
          5 print(hash(b))
          6 a = b
          7 print(hash(a))
```

```
2570
2570
2570
2570
```

```
In [160]: 1 print(hash(25))
          2 a = 25
          3 print(hash(a))
          4 b = 25
          5 print(hash(b))
          6 a = b
          7 print(hash(a))
```

```
25
25
25
25
```

```
In [161]: 1 print(hash(2.5))
          2 a = 2.5
          3 print(hash(a))
          4 b = 2.5
          5 print(hash(b))
          6 print(hex(hash(b)))
          7 a = b
          8 print(len(hex(hash(a))))
          9 print(hash(b))
```

```
1152921504606846978
1152921504606846978
1152921504606846978
0x10000000000000002
18
1152921504606846978
```

```
In [162]: 1 for i in range(5):
          2     for j in range(4):
          3         pass
          4 print(i, j)
```

4 3

```
In [163]: 1
          2
          3
          4 s1 = """
          5 "'And' and 'or' are the basic operations of logic. Together with 'no' (the
          6 logical operation
          7 of negation) they are a complete set of basic logical operations – all other
          8 logical
          9 operations, no matter how complex, can be obtained by suitable combinations of
          0 these."
          1 — John von Neumann, The Computer and the Brain
          2 """
          3
          4 s2 = """
          5 Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor
          6 incididunt ut
          7 labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud
          8 exercitation ullamco laboris
          9 nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in
          0 reprehenderit in voluptate velit
          1 esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat
          2 non proident, sunt
          3 in culpa qui officia deserunt mollit anim id est laborum.
          4 """
          5
          6 charCounts = {}
          7 for c in s1:
          8     if (ord(c) >= ord("A") and ord(c) <= ord("Z")) or (ord(c) >= ord("a")
          9 and ord(c) <= ord("z")):
          0         charCounts[c] = charCounts.get(c, 0) + 1
          1 print(charCounts)
```

```
{'A': 1, 'n': 20, 'd': 4, 'a': 23, 'o': 31, 'r': 12, 'e': 27, 't': 21, 'h': 11, 'b': 7, 's': 9, 'i': 17, 'c': 10, 'p': 7, 'f': 4, 'l': 12, 'g': 6, 'T': 2, 'w': 2, 'y': 2, 'm': 6, 'x': 1, 'u': 3, 'J': 1, 'v': 1, 'N': 1, 'C': 1, 'B': 1}
```

```
In [164]: 1 d1 = {"a": 10, "b": 20, "c": 30}
2 d2 = {"d": 100, "e": 200, "f": 300}
3 d3 = {"f": 30, "g": 40}
4 keysList, valuesList = [], []
5 for d in d1, d2, d3:
6     for k, v in d.items():
7         keysList.append(k)
8         valuesList.append(v)
9 print(keysList)
10 print(valuesList)
```

```
['a', 'b', 'c', 'd', 'e', 'f', 'f', 'g']
[10, 20, 30, 100, 200, 300, 30, 40]
```

```
In [165]: 1 keysList, valuesList = [], []
2 keysList = list(d1.keys())
3 keysList.extend(list(d2.keys()))
4 keysList.extend(list(d3.keys()))
5 print(keysList)
```

```
['a', 'b', 'c', 'd', 'e', 'f', 'f', 'g']
```

```
In [166]: 1 valuesList = list(d1.values())
2 valuesList.extend(list(d2.values()))
3 valuesList.extend(list(d3.values()))
4 print(valuesList)
```

```
[10, 20, 30, 100, 200, 300, 30, 40]
```

```
In [167]: 1 print(list(d1.keys()) + list(d2.keys()) + list(d3.keys()))
2 print(list(d1.values()) + list(d2.values()) + list(d3.values()))
```

```
['a', 'b', 'c', 'd', 'e', 'f', 'f', 'g']
[10, 20, 30, 100, 200, 300, 30, 40]
```

```
In [168]: 1 grades = {"John": [90, 95, 98], "Eric": [86, 84, 92], "Michael": [90, 89, 85]}
2 exam = {"Eric": 99, "John": 100}
3 for k, v in grades.items():
4     if k in exam:
5         v.insert(0, exam[k])
6     else:
7         v.insert(0, None)
8 print(grades)
```

```
{'John': [100, 90, 95, 98], 'Eric': [99, 86, 84, 92], 'Michael': [None, 90, 89, 85]}
```

```
In [169]: 1 for k, v in grades.items():
2     score = exam.get(k)
3     grades[k] = [score] + grades[k]
4 print(grades)
```

```
{'John': [100, 100, 90, 95, 98], 'Eric': [99, 99, 86, 84, 92], 'Michael': [None, None, 90, 89, 85]}
```

```
In [170]: 1 for k, v in grades.items():
          2     v.insert(0, exam.get(k))
          3 print(grades)
```

```
{'John': [100, 100, 100, 90, 95, 98], 'Eric': [99, 99, 99, 86, 84, 92], 'Michael': [None, None,
None, 90, 89, 85]}
```

```
In [171]: 1 for k in grades:
          2     print(k)
```

```
John
Eric
Michael
```

```
In [172]: 1 l = ["AAPL", "AAPL", "Aapl", "aapl", "MSFT"]
          2 print(list(set(l)))
```

```
['Aapl', 'AAPL', 'aapl', 'MSFT']
```

```
In [173]: 1 l = [e.casefold() for e in l]
          2 print(list(set(l)))
```

```
['msft', 'aapl']
```

```
In [174]: 1 se = {e.casefold() for e in l}
          2 print(se)
```

```
{'msft', 'aapl'}
```

```
In [175]: 1 data = {"d1": {"a": 1, "b": 2, "c": 3}, "d2": {"b": 20, "c": 30, "d": 40},
          2          "d3": {"d": 100, "x": 200}}
          3 uniqueKeys = {k for v in (data["d1"], data["d2"], data["d3"]) for k in v}
          print(uniqueKeys)
```

```
{'x', 'd', 'b', 'a', 'c'}
```

```
In [176]: 1 try:
          2     s1 = {[0, 1]}
          3     print(s1)
          4 except TypeError as ex:
          5     print(type(ex).__name__, ": ", *ex.args, ".", sep = " ")
```

```
TypeError: cannot use 'list' as a set element (unhashable type: 'list').
```

```
In [177]: 1 m = [0] * 3
          2 print(m)
          3 print(m[0] is m[1])
```

```
[0, 0, 0]
True
```

```
In [178]: 1 m = [[0] * 3 for _ in range(3)]
          2 print(m)
          3 print(m[0] is m[1])
          4 print(m[0][0] is m[0][1])
```

```
[[0, 0, 0], [0, 0, 0], [0, 0, 0]]
```

```
False
```

```
True
```

```
In [179]: 1 m = [[10000 for i in range(3)] for _ in range(4)]
          2 print(m)
```

```
[[10000, 10000, 10000], [10000, 10000, 10000], [10000, 10000, 10000], [10000, 10000, 10000]]
```

```
In [180]: 1 pprint.pprint(m, width = 1)
```

```
[[10000,
  10000,
  10000],
 [10000,
  10000,
  10000],
 [10000,
  10000,
  10000],
 [10000,
  10000,
  10000]]
```

```
In [181]: 1 print(m[0] is m[1])
          2 print(m[0][0] is m[0][1])
          3 print(hash(m[0][0]), hash(m[0][1]))
```

```
False
```

```
True
```

```
10000 10000
```

```
In [182]: 1 print(id(m[0][0]), id(m[0][1]))
```

```
125017713565616 125017713565616
```

```
In [183]: 1 m[0][0] = 1001
          2 pprint.pprint(m)
```

```
[[1001, 10000, 10000],
 [10000, 10000, 10000],
 [10000, 10000, 10000],
 [10000, 10000, 10000]]
```

```
In [184]: 1 print(id(m[0][0]), id(m[0][1]))
          2 print(m[0][0] is m[0][1])
```

```
125017714245296 125017713565616
```

```
False
```

```
In [185]: data = [{"open": 100, "high": 120, "low": 90, "close": 110}, {"open": 110,
1 "high": 130, "low": 80, "close": 120}, {"open": 120, "high": 140, "low": 70,
2 "close": 130}, {"open": 130, "high": 150, "low": 60, "close": 140},]
3 ranges = []
4 for d in data:
5     ranges.append(d["high"] - d["low"])
print(ranges)
```

```
[30, 50, 70, 90]
```

In [186]:

```
1 data = [{"open": 100, "high": 120, "low": 90, "close": 110}, {"open": 110,
2 "high": 130, "low": 80, "close": 120}, {"open": 120, "high": 140, "low": 70,
3 "close": 130}, {"open": 130, "high": 150, "low": 60, "close": 140},]
ranges = [d["high"] - d["low"] for d in data]
print(ranges)
```

```
[30, 50, 70, 90]
```

In [187]:

```
1 bogusComposites = {i for i in range(1, 101) for j in range(2, 10) if i % j ==
2 0}
3 allNumbers = {i for i in range(11, 101)} | {1}
print(allNumbers - bogusComposites)
```

```
{1, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97}
```

In [188]:

```
1 allNumbers = set(range(1, 101))
2 print(allNumbers - bogusComposites)
```

```
{1, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97}
```

In [189]:

```
1 data = [{"symbol": "ABCD", "name": "ABCD Company", "ranking": 2, "risk": 0.2},
2 {"symbol": "BCDE", "name": "BCDE Company", "ranking": 5, "risk": 0.2},
3 {"symbol": "CDEF", "name": "CDEF Company", "ranking": 8, "risk": 0.5},
{"symbol": "DEFG", "name": "DEFG Company", "ranking": 7, "risk": 0.8},
{"symbol": "EFGH", "name": "EFGH Company", "ranking": 9, "risk": 0.6},
{"symbol": "FGHI", "name": "FGHI Company", "ranking": 10, "risk": 0.1},
{"symbol": "GHIJ", "name": "GHIJ Company", "ranking": 3, "risk": 0.6},
{"symbol": "HIJK", "name": "HIJK Company", "ranking": 5, "risk": 0.5},
{"symbol": "IJKL", "name": "IJKL Company", "ranking": 5, "risk": 0.7},
{"symbol": "JKLM", "name": "JKLM Company", "ranking": 6, "risk": 0.9},
{"symbol": "KLMN", "name": "KLMN Company", "ranking": 6, "risk": 0.4},
{"symbol": "LMNO", "name": "LMNO Company", "ranking": 8, "risk": 0.4},
{"symbol": "MNOP", "name": "MNOP Company", "ranking": 8, "risk": 0.2},
{"symbol": "NOPQ", "name": "NOPQ Company", "ranking": 1, "risk": 0.5},
{"symbol": "OPQR", "name": "OPQR Company", "ranking": 9, "risk": 0.2},
{"symbol": "PQRS", "name": "PQRS Company", "ranking": 10, "risk": 0.9},
{"symbol": "QRST", "name": "QRST Company", "ranking": 3, "risk": 0.4},
{"symbol": "RSTU", "name": "RSTU Company", "ranking": 7, "risk": 0.3},
{"symbol": "STUV", "name": "STUV Company", "ranking": 8, "risk": 0.1},
{"symbol": "TUVW", "name": "TUVW Company", "ranking": 7, "risk": 0.9}]
filtered = [d for d in data if d["ranking"] > 5 and d["risk"] < 0.6]
print(filtered)
```

```
[{'symbol': 'CDEF', 'name': 'CDEF Company', 'ranking': 8, 'risk': 0.5}, {'symbol': 'FGHI', 'name':
'FGHI Company', 'ranking': 10, 'risk': 0.1}, {'symbol': 'KLMN', 'name': 'KLMN Company', 'ranking':
6, 'risk': 0.4}, {'symbol': 'LMNO', 'name': 'LMNO Company', 'ranking': 8, 'risk': 0.4}, {'symbol':
'MNOP', 'name': 'MNOP Company', 'ranking': 8, 'risk': 0.2}, {'symbol': 'OPQR', 'name': 'OPQR
Company', 'ranking': 9, 'risk': 0.2}, {'symbol': 'RSTU', 'name': 'RSTU Company', 'ranking': 7,
'risk': 0.3}, {'symbol': 'STUV', 'name': 'STUV Company', 'ranking': 8, 'risk': 0.1}]
```

In [190]:

```
1 for d in filtered:
2     d["weight"] = float(f"{d["ranking"] * d["risk"]:.2f}")
```

```
3 print(filtered)
```

```
[{'symbol': 'CDEF', 'name': 'CDEF Company', 'ranking': 8, 'risk': 0.5, 'weight': 4.0}, {'symbol': 'FGHI', 'name': 'FGHI Company', 'ranking': 10, 'risk': 0.1, 'weight': 1.0}, {'symbol': 'KLMN', 'name': 'KLMN Company', 'ranking': 6, 'risk': 0.4, 'weight': 2.4}, {'symbol': 'LMNO', 'name': 'LMNO Company', 'ranking': 8, 'risk': 0.4, 'weight': 3.2}, {'symbol': 'MNOP', 'name': 'MNOP Company', 'ranking': 8, 'risk': 0.2, 'weight': 1.6}, {'symbol': 'OPQR', 'name': 'OPQR Company', 'ranking': 9, 'risk': 0.2, 'weight': 1.8}, {'symbol': 'RSTU', 'name': 'RSTU Company', 'ranking': 7, 'risk': 0.3, 'weight': 2.1}, {'symbol': 'STUV', 'name': 'STUV Company', 'ranking': 8, 'risk': 0.1, 'weight': 0.8}]
```

```
In [191]: 1 weightedStocks = [dict(symbol = d["symbol"], weight = d["weight"]) for d in
          2 print(weightedStocks)
```

```
[{'symbol': 'CDEF', 'weight': 4.0}, {'symbol': 'FGHI', 'weight': 1.0}, {'symbol': 'KLMN', 'weight': 2.4}, {'symbol': 'LMNO', 'weight': 3.2}, {'symbol': 'MNOP', 'weight': 1.6}, {'symbol': 'OPQR', 'weight': 1.8}, {'symbol': 'RSTU', 'weight': 2.1}, {'symbol': 'STUV', 'weight': 0.8}]
```

```
In [192]: 1 weightedStocks = [dict(symbol = d["symbol"], weight = round(d["ranking"] *
          2 d["risk"], 2)) for d in data if d["ranking"] >= 5 and d["risk"] < 0.6]
          print(weightedStocks)
```

```
[{'symbol': 'BCDE', 'weight': 1.0}, {'symbol': 'CDEF', 'weight': 4.0}, {'symbol': 'FGHI', 'weight': 1.0}, {'symbol': 'HIJK', 'weight': 2.5}, {'symbol': 'KLMN', 'weight': 2.4}, {'symbol': 'LMNO', 'weight': 3.2}, {'symbol': 'MNOP', 'weight': 1.6}, {'symbol': 'OPQR', 'weight': 1.8}, {'symbol': 'RSTU', 'weight': 2.1}, {'symbol': 'STUV', 'weight': 0.8}]
```

```
In [193]: 1 weightedStocks = [{"symbol": d["symbol"], "weight": round(d["ranking"] *
          2 d["risk"], 2)} for d in data if d["ranking"] >= 5 and d["risk"] < 0.6]
          print(weightedStocks)
```

```
[{'symbol': 'BCDE', 'weight': 1.0}, {'symbol': 'CDEF', 'weight': 4.0}, {'symbol': 'FGHI', 'weight': 1.0}, {'symbol': 'HIJK', 'weight': 2.5}, {'symbol': 'KLMN', 'weight': 2.4}, {'symbol': 'LMNO', 'weight': 3.2}, {'symbol': 'MNOP', 'weight': 1.6}, {'symbol': 'OPQR', 'weight': 1.8}, {'symbol': 'RSTU', 'weight': 2.1}, {'symbol': 'STUV', 'weight': 0.8}]
```

```
In [194]: 1 m = [[0] * 2] * 3
          2 print(m)
          3 for r in range(len(m)):
          4     for c in range(len(m[r])):
          5         print(id(m[r][c]))
```

```
[[0, 0], [0, 0], [0, 0]]
11370000
11370000
11370000
11370000
11370000
11370000
```

```
In [195]: 1 m = [{0}] * 2 * 3
          2 print(m)
          3 for r in range(len(m)):
          4     for c in range(len(m[r])):
          5         print(id(m[r][c]))
```

```
[[{0}, {0}], [{0}, {0}], [{0}, {0}]]
125017713411456
125017713411456
125017713411456
125017713411456
125017713411456
125017713411456
```

```
In [196]: 1 m[0][0] = 1
          2 for r in range(len(m)):
          3     for c in range(len(m[r])):
          4         print(id(m[r][c]))
          5 print(m)
```

```
11370032
125017713411456
11370032
125017713411456
11370032
125017713411456
[[1, {0}], [1, {0}], [1, {0}]]
```

```
In [197]: 1 m = [{0} for _ in range(2)] for _ in range(3)]
          2 print(m)
          3 for r in range(len(m)):
          4     for c in range(len(m[r])):
          5         print(id(m[r][c]))
          6 m[0][0] = 1
          7 print(m)
```

```
[[{0}, {0}], [{0}, {0}], [{0}, {0}]]
125017713413696
125017713410112
125017713415040
125017713411904
125017713412576
125017713412800
[[1, {0}], [{0}, {0}], [{0}, {0}]]
```

```
In [198]: 1 m = [{0} for _ in range(2)] for _ in range(3)]
          2 print(m)
          3 for r in range(len(m)):
          4     for c in range(len(m[r])):
          5         print(id(m[r][c]))
          6 m[0][0] = {1}
          7 print(m)
```

```
[[{0}, {0}], [{0}, {0}], [{0}, {0}]]
125017713413696
125017713411456
125017713415488
125017713414816
125017713414592
125017713416160
[[{1}, {0}], [{0}, {0}], [{0}, {0}]]
```

```
In [199]: 1 m = [[0 for _ in range(2)] for _ in range(3)]
          2 for r in range(len(m)):
          3     for c in range(len(m[r])):
          4         print(id(m[r][c]))
```

```
11370000
11370000
11370000
11370000
11370000
11370000
```

```
In [200]: 1 m = [[10000 for _ in range(2)] for _ in range(3)]
          2 for r in range(len(m)):
          3     for c in range(len(m[r])):
          4         print(id(m[r][c]))
          5 m[0][0] = 1
          6 print(m)
```

```
125017713567216
125017713567216
125017713567216
125017713567216
125017713567216
125017713567216
[[1, 10000], [10000, 10000], [10000, 10000]]
```

```
In [201]: 1 m = [[{0} for _ in range(2)] for _ in range(3)]
          2 for r in range(len(m)):
          3     for c in range(len(m[r])):
          4         print(id(m[r][c]))
```

```
125017713410112
125017713415488
125017713414144
125017713411904
125017713416160
125017713414368
```

```
In [202]: 1 # print(input("Text box displayed ?"))
```

```
In [203]: 1 a = 0
          2 b = complex(real = 3.0, imag = 4)
          3 print(repr(a))
          4 print(repr(b))
          5 print(b)
```

```
0
(3+4j)
(3+4j)
```

```
In [204]: 1 values = [4, 3, 1, 2]
          2 try:
          3     minimum = abs(values[0])
          4     for value in values[1:]:
```

```

5         if abs(value) < minimum:
6             minimum = abs(value)
7     except IndexError:
8         minimum = 0
9         pass
10    finally:
11        print(f"Minimum is: {minimum}")

```

Minimum is: 1

```

In [205]: 1 values = []
2
3     try:
4         minimum = abs(values[0])
5         for value in values[1:]:
6             if abs(value) < minimum:
7                 minimum = abs(value)
8     except IndexError:
9         minimum = 0
10        pass
11    finally:
12        print(f"Minimum is: {minimum}")

```

Minimum is: 0

```

In [206]: 1 try:
2         raise ValueError("Custom message")
3     except ValueError as ex:
4         print(type(ex).      + ": " + ex.args[0] + ".")

```

ValueError: Custom message.

```

In [207]: 1 l = [4, 3, 2, 1, 0]
2     iterator = iter(l)
3     try:
4         while True:
5             print(next(iterator))
6     except StopIteration as ex:
7         print(type(ex).      , ": ", *ex.args, ".", sep = "")

```

```

4
3
2
1
0
StopIteration: .

```

```

In [208]: 1 try:
2         raise IndexError
3     except IndexError as ex:
4         print(type(ex).      , ": ", *ex.args, ".", sep = "")

```

IndexError: .

```
In [209]: 1 rangeObject = range(10)
          2 rangeIterator = iter(rangeObject)
          3 print(next(rangeIterator))
          4 print(next(rangeIterator))
          5 print(next(rangeIterator))
          6 print(list(rangeIterator))
```

```
0
1
2
[3, 4, 5, 6, 7, 8, 9]
```

```
In [210]: 1 rangeObj = range(5)
          2 print(list(rangeObj))
          3 print(list(rangeObj))
```

```
[0, 1, 2, 3, 4]
[0, 1, 2, 3, 4]
```

```
In [211]: 1 enumIterator = enumerate(string.ascii_lowercase)
          2 print(enumIterator)
          3 l = list(enumIterator)
          4 print(l)
          5 l = list(enumIterator)
          6 print(dir(type(iter([0]))))
          7 print(l)
          8 try:
          9     print(next(enumIterator))
         10 except StopIteration as ex:
         11     print(type(ex).      , ": ", *ex.args, ".", sep = "")
```

```
<enumerate object at 0x71b3f4718ef0>
[(0, 'a'), (1, 'b'), (2, 'c'), (3, 'd'), (4, 'e'), (5, 'f'), (6, 'g'), (7, 'h'), (8, 'i'), (9, 'j'),
(10, 'k'), (11, 'l'), (12, 'm'), (13, 'n'), (14, 'o'), (15, 'p'), (16, 'q'), (17, 'r'), (18, 's'),
(19, 't'), (20, 'u'), (21, 'v'), (22, 'w'), (23, 'x'), (24, 'y'), (25, 'z')]
['_class_', '__delattr__', '__dir__', '__doc__', '__eq__', '__format__', '__ge__',
'__getattr__', '__getstate__', '__gt__', '__hash__', '__init__', '__init_subclass__',
'__iter__', '__le__', '__length_hint__', '__lt__', '__ne__', '__new__', '__next__', '__reduce__',
'__reduce_ex__', '__repr__', '__setattr__', '__setstate__', '__sizeof__', '__str__',
'__subclasshook__']
[]
StopIteration: .
```

```
In [212]: 1 print(type(enumIterator))
          2 print(type(iter(enumIterator)))
          3 print(issubclass(type(iter([0])), type(enumIterator)))
```

```
<class 'enumerate'>
<class 'enumerate'>
False
```

```
In [213]: 1 print(iter(enumIterator) is enumIterator)
```

```
True
```

```
In [214]: 1 print(type(iter({0, 1, 2})))
```

```
<class 'set_iterator'>
```

```
In [215]: 1 print(type(iter({0: "0", 1: "1", "2": 2})))
```

```
<class 'dict_keyiterator'>
```

```
In [216]: 1 l = [4, 3, 2, 1, 0]
2 iterator = iter(l)
3 for i in iterator:
4     print(i)
```

```
4
3
2
1
0
```

```
In [217]: 1 try:
2         print(next(l))
3 except TypeError as ex:
4         print(type(ex).      , ": ", *ex.args, ".", sep = "")
```

```
TypeError: 'list' object is not an iterator.
```

```
In [218]: 1 start = time.perf_counter()
2 squaresGenerator = (i ** 10 for i in range(1_000_000))
3 end = time.perf_counter()
4 print(f"Elapsed: {end - start}")
```

```
Elapsed: 8.376799814868718e-05
```

```
In [219]: 1 start = time.perf_counter()
2 squaresListComp = [i ** 10 for i in range(1_000_000)]
3 end = time.perf_counter()
4 print(f"Elapsed: {end - start}")
```

```
Elapsed: 0.22543958500318695
```

```
In [220]: 1 print((7.16e-5 / 1.824e-1) ** -1)
2 print(105 / 9)
3 print(10.5e1 / 9e0)
4 print(99 / 1)
5 print(9.9e1 / 1e0) # 1 order of magnitude difference means a multiple of
                    between 10 and 100
```

```
2547.486033519553
11.666666666666666
11.666666666666666
99.0
99.0
```

```
In [221]: 1 print(999_999 ** 10)
```

```
9999900000044999880000209999748000209999880000044999990000001
```

```
In [222]: 1 print(squaresGenerator is iter(squaresGenerator))
```

```
True
```

```
In [223]: 1 l = [10, "abc", 3.14, True]
          2 for i, e in enumerate(l):
          3     print(f"{i}: {e}")
```

```
0: 10
1: abc
2: 3.14
3: True
```

```
In [224]: 1 def f(a, *b, x, **y):
          2     print("Positional/Named:", a)
          3     print("Variable positional:", b)
          4     print("Named only:", x)
          5     print("Variable named:", y)
          6
          7
          8 f(0, 1, 2, x = 3, y = 4, z = 5)
```

```
Positional/Named: 0
Variable positional: (1, 2)
Named only: 3
Variable named: {'y': 4, 'z': 5}
```

```
In [225]: 1 # f(10, 2, x = 10, 20)
```

```
In [226]: 1 f(10, 2, x = 10, y = 20)
```

```
Positional/Named: 10
Variable positional: (2,)
Named only: 10
Variable named: {'y': 20}
```

```
In [227]: 1 f(10, z = 2, x = 10, y = 20)
```

```
Positional/Named: 10
Variable positional: ()
Named only: 10
Variable named: {'z': 2, 'y': 20}
```

```
In [228]: 1 f(10, x = 2, y = 10, z = 20)
```

```
Positional/Named: 10
Variable positional: ()
Named only: 2
Variable named: {'y': 10, 'z': 20}
```

```
In [229]: 1 # f(a = 10, 2, x = 10, y = 20)
```

```
In [230]: 1 f(x = 10, y = 20, z = 30, a = 10)
```

```
Positional/Named: 10
Variable positional: ()
Named only: 10
Variable named: {'y': 20, 'z': 30}
```

```
In [231]: 1 try:
          2     f(10, a = 2, x = 10, y = 20)
          3 except TypeError as ex:
          4     print(type(ex).      )
```

```
TypeError
```

```
In [232]: 1 def f1(a, *, x, **y):
          2     print(f"a: {a}, x: {x}, y: {y}")
```

```
In [233]: 1 try:
          2     f1(10, a = 20, x = 2, y = 10)
          3 except TypeError as ex:
          4     print(type(ex).      )
```

```
TypeError
```

```
In [234]: 1 f1(a = 1, x = 2)
```

```
a: 1, x: 2, y: {}
```

```
In [235]: 1 print(max("cba"))
```

```
c
```

```
In [236]: 1 def f2(a, b):
          2     print(f"a: {a}; b: {b}")
          3
          4
          5 f2(2, 3)
```

```
a: 2; b: 3
```

```
In [237]: 1 f2(2, b = 3)
```

```
a: 2; b: 3
```

```
In [238]: 1 # f2(a = 2, 3)
```

```
In [239]: 1 f2(b = 1, a = 0)
```

```
a: 0; b: 1
```

```
In [240]: 1 # def f3(a = 0, b):
          2 #     print(f"a: {a}; b: {b}")
```

```
In [241]: 1 def f3(a, b = 0):
          2     print(f"a: {a}; b: {b}")
          3
```

```
4
5 f3(-1, 1)
```

```
a: -1; b: 1
```

```
In [242]: 1 # f3(a = -1, 1)
```

```
In [243]: 1 f3(a = -1, b = 1)
```

```
a: -1; b: 1
```

```
In [244]: 1 def f4(a, b = 1, *, x):
2         print(a, b, x)
3
4
5 f4(0, 2, x = 3)
```

```
0 2 3
```

```
In [245]: 1 def f5(a, b = 1, *, x = 2, y):
2         print(a, b, x, y)
3
4
5 f5(0, 2, x = 3, y = 4)
```

```
0 2 3 4
```

```
In [246]: 1 def f6(**x):
2         print(x)
3
4
5 f6(a = 1, b = 2, c = 3)
```

```
{'a': 1, 'b': 2, 'c': 3}
```

```
In [247]: 1 # def f7(*, **x):
2         #     print(x)
3         #
4         #
5         # f7(a = 1, b = 2)
```

```
In [248]: 1 def f8(a, *, b):
2         print(f"a: {a}; b: {b}")
3
4
5 f8(b = 1, a = 0)
```

```
a: 0; b: 1
```

```
In [249]: 1 def toJSON(a, *, b, **c):
          2     print(a, b, c)
          3
          4
          5 toJSON(0, b = 3, c1 = 0, c2 = 1)
```

```
0 3 {'c1': 0, 'c2': 1}
```

```
In [250]: 1 def avg(a, *b):
          2     return (sum(b) + a) / (len(b) + 1)
          3
          4
          5 print(avg(0, 1, 2, 3, 4))
          6 print(avg(0))
          7 print(avg(0, 1))
```

```
2.0
0.0
0.5
```

```
In [251]: 1 def separator(b = 10, *, a = "-"):
          2     return a * b
          3
          4
          5 print(separator())
          6 print(separator(a = "A"))
          7 print(separator(5))
          8 print(separator(a = "B", b = 5))
          9 print(separator(5, a = "B"))
```

```
-----
AAAAAAAAAA
-----
BBBBBB
BBBBBB
```

```
In [252]: 1 print(sep = "|", *("AB", "CD"))
```

```
AB|CD
```

```
In [253]: 1 # def f(a, *b, c):
          2 #     print(a, b, c)
          3 #
          4 #
          5 # f(c = 2, 0, 1, 2)
```

```
In [254]: 1 # def f(*a, b = 0):
          2 #     print(a, b)
          3 #
          4 #
          5 # f(b = 1, 0, 2)
```

```
In [255]: 1 try:
          2     f(b = 1, *(0, 2))
          3 except TypeError as ex:
          4     print(type(ex).      )
```

TypeError

```
In [256]: 1 # def f(a, *, b):
          2 #     print(a, b)
          3 #
          4 #
          5 # f(b = 1, 0)
```

```
In [257]: 1 # def f(a, b):
          2 #     print(a, b)
          3 #
          4 #
          5 # f(b = 1, 0)
```

```
In [258]: 1 # def f(a, b = 2):
          2 #     print(a, b)
          3 #
          4 #
          5 # f(b = 1, 0)
```

```
In [259]: 1 def f(a, b = 2):
          2     print(a, b)
          3
          4
          5 f(b = 1, *(0,))
```

0 1

```
In [260]: 1 anonymous = lambda iterable: len(set(iterable))
          2 print(anonymous("abcd"))
          3 print(anonymous("abab"))
          4 try:
          5     print(anonymous())
          6 except TypeError as ex:
          7     print(type(ex).      )
          8 print(anonymous(""))
          9 print(anonymous([0, 1, 2]))
         10 print(anonymous([0]))
         11 print(anonymous([]))
         12 print(anonymous([1, 1, 2, 2, 3, 4]))
```

```
4
2
TypeError
0
3
1
0
4
```

In [261]:

```
1 s = "This is the first sentence. This is the scecond sentence. This is not the
2 fourth sentence, it is the third sentence."
3
4
5 def f(s = ""):
6     return dict(collections.Counter(s))
7
8 print(f(s))
9 print(f(""))
```

```
{'T': 3, 'h': 9, 'i': 10, 's': 13, ' ': 20, 't': 13, 'e': 17, 'f': 2, 'r': 3, 'n': 10, 'c': 6, 'l': 3, 'o': 3, 'd': 2, 'u': 1, ',': 1}
{}
```

In [262]:

```
1
2
3
4 s = "This is the first sentence. This is the scecond sentence. This is not the
5 fourth sentence, it is the third sentence."
6
7 def f(s = ""):
8     words = re.split("\\.| |", s)
9     for word in words:
10         if word == "":
11             words.remove(word)
12     frequencies = dict.fromkeys(set(words), 0)
13     for word in words:
14         frequencies[word] += 1
15     print(frequencies)
16
17 f(s)
18
19
20
21
22
23
24
25
```

```
{'sentence': 4, 'it': 1, 'This': 3, 'third': 1, 'the': 4, 'second': 1, 'is': 4, 'first': 1, 'fourth': 1, 'not': 1}
```

```
In [263]: 1 def f(s = ""):
2           s = s.replace(".", "")
3           s = s.replace(",", "")
4           words = s.split()
5           return dict(collections.Counter(words))
6
7
8 print(f(s))
```

```
In [264]: 1 print(round(1.812))
```

2

```
In [265]: 1 n, d = (0.3).as_integer_ratio()
          2 print(n, d)
          3 print(1 / 3)
```

5404319552844595 18014398509481984
0.3333333333333333

```
In [266]: 1 print(round(0.325, 2))
```

0.33

```
In [267]: 1 print(f"{0.325:.40f}")
```

0.325000000000000000111022302462515654042363

```
In [268]: 1 def toJSON(a, *, b, **c):
2           print(a, b, c)
3
4
5 try:
6     toJSON(0, 1, 2, b = 3, c1 = 4, c2 = 5)
7 except TypeError as ex:
8     print(type(ex).      )
```

TypeError

```
In [269]: 1 def toJSON(a, *, b, **c):
           2     print(a, b, c)
           3
           4
           5 toJSON(0, b = 3, c1 = 4, c2 = 5)
```

```
0 3 {'c1': 4, 'c2': 5}
```

```
In [270]: 1 # def toJSON(a, *):
          2 #     print(a)
```

```
3 #
4 #
5 # toJSON(0, 1, 2)
```

```
In [271]: 1 def toJSON(a, *args):
          2     print(a, args)
          3
          4
          5 toJSON(0, 1, 2)
```

```
0 (1, 2)
```

```
In [272]: 1 l = ["AB CD", "EF CC", "GH BC", "IJ AC", "KL CA", "MN JQ", "OP BA"]
          2 print(sorted(l, key = lambda e: (e[3], e[4])))
```

```
['IJ AC', 'OP BA', 'GH BC', 'KL CA', 'EF CC', 'AB CD', 'MN JQ']
```

```
In [273]: 1 # print(sorted(l, key = lambda e: e[len(e) - i] for i in range(len(e))))
          2 print(sorted(l, key = lambda e: e[len(e) - 1]))
```

```
['KL CA', 'OP BA', 'EF CC', 'GH BC', 'IJ AC', 'AB CD', 'MN JQ']
```

```
In [274]: 1 a = (1 for _ in range(5))
          2 print(a)
          3 print(lambda _: "ABCD")
          4 print(sorted)
          5 print(range)
```

```
<generator object <genexpr> at 0x71b3f469d220>
<function <lambda> at 0x71b3f4643320>
<built-in function sorted>
<class 'range'>
```

```
In [275]: 1 def f(*a, b):
          2     print(a, b)
          3
          4
          5 f(b = 0, *(1, 2))
```

```
(1, 2) 0
```

```
In [276]: 1 def f(a, *b, c):
          2     print(a, b, c)
          3
          4
          5 f(0, c = 1, *(2, 3))
```

```
0 (2, 3) 1
```

```
In [277]: 1 print(statistics.fmean)
```

```
<function fmean at 0x71b438c06da0>
```

```
In [278]: 1 print(dict([("a", 97), ("b", 98), ("c", 99)]))
```

```
{'a': 97, 'b': 98, 'c': 99}
```

```
In [279]: 1 widgets = [f"w{i}" for i in range(1, 21)]
          2 skus = [f"sku{i}" for i in range(1, len(widgets) + 1)]
          3 print(widgets)
          4 print(skus)
```

```
['w1', 'w2', 'w3', 'w4', 'w5', 'w6', 'w7', 'w8', 'w9', 'w10', 'w11', 'w12', 'w13', 'w14', 'w15',
 'w16', 'w17', 'w18', 'w19', 'w20']
['sku1', 'sku2', 'sku3', 'sku4', 'sku5', 'sku6', 'sku7', 'sku8', 'sku9', 'sku10', 'sku11', 'sku12',
 'sku13', 'sku14', 'sku15', 'sku16', 'sku17', 'sku18', 'sku19', 'sku20']
```

```
In [280]: 1 print({w: s for w, s in zip(widgets, skus)})
```

```
{'w1': 'sku1', 'w2': 'sku2', 'w3': 'sku3', 'w4': 'sku4', 'w5': 'sku5', 'w6': 'sku6', 'w7': 'sku7',
 'w8': 'sku8', 'w9': 'sku9', 'w10': 'sku10', 'w11': 'sku11', 'w12': 'sku12', 'w13': 'sku13', 'w14':
 'sku14', 'w15': 'sku15', 'w16': 'sku16', 'w17': 'sku17', 'w18': 'sku18', 'w19': 'sku19', 'w20':
 'sku20'}
```

```
In [281]: 1 print(dict(zip(widgets, skus)))
```

```
{'w1': 'sku1', 'w2': 'sku2', 'w3': 'sku3', 'w4': 'sku4', 'w5': 'sku5', 'w6': 'sku6', 'w7': 'sku7',
 'w8': 'sku8', 'w9': 'sku9', 'w10': 'sku10', 'w11': 'sku11', 'w12': 'sku12', 'w13': 'sku13', 'w14':
 'sku14', 'w15': 'sku15', 'w16': 'sku16', 'w17': 'sku17', 'w18': 'sku18', 'w19': 'sku19', 'w20':
 'sku20'}
```

```
In [282]: 1 suits = "shdc" # Spades, Hearts, Diamonds, Clubs
          2 ranks = list("23456789") + ["10", "J", "Q", "K", "A"]
          3 print(suits)
          4 print(ranks)
```

```
shdc
['2', '3', '4', '5', '6', '7', '8', '9', '10', 'J', 'Q', 'K', 'A']
```

```
In [283]: 1 print([(rank, suit) for rank in ranks] for suit in suits))
```

```
[('2', 's'), ('3', 's'), ('4', 's'), ('5', 's'), ('6', 's'), ('7', 's'), ('8', 's'), ('9', 's'),
 ('10', 's'), ('J', 's'), ('Q', 's'), ('K', 's'), ('A', 's')], [('2', 'h'), ('3', 'h'), ('4', 'h'),
 ('5', 'h'), ('6', 'h'), ('7', 'h'), ('8', 'h'), ('9', 'h'), ('10', 'h'), ('J', 'h'), ('Q', 'h'),
 ('K', 'h'), ('A', 'h')], [('2', 'd'), ('3', 'd'), ('4', 'd'), ('5', 'd'), ('6', 'd'), ('7', 'd'),
 ('8', 'd'), ('9', 'd'), ('10', 'd'), ('J', 'd'), ('Q', 'd'), ('K', 'd'), ('A', 'd')], [('2', 'c'),
 ('3', 'c'), ('4', 'c'), ('5', 'c'), ('6', 'c'), ('7', 'c'), ('8', 'c'), ('9', 'c'), ('10', 'c'),
 ('J', 'c'), ('Q', 'c'), ('K', 'c'), ('A', 'c')]]
```

```
In [284]: 1 pprint.pprint([(rank, suit) for rank in ranks] for suit in suits))
```

```
[('2', 's'),
 ('3', 's'),
 ('4', 's'),
 ('5', 's'),
 ('6', 's'),
 ('7', 's'),
 ('8', 's'),
 ('9', 's'),
 ('10', 's'),
 ('J', 's'),
 ('Q', 's'),
 ('K', 's'),
 ('A', 's')],
 [('2', 'h'),
 ('3', 'h'),
 ('4', 'h'),
 ('5', 'h'),
 ('6', 'h'),
 ('7', 'h'),
 ('8', 'h'),
 ('9', 'h'),
 ('10', 'h'),
 ('J', 'h'),
 ('Q', 'h'),
 ('K', 'h'),
 ('A', 'h')],
 [('2', 'd'),
 ('3', 'd'),
 ('4', 'd'),
 ('5', 'd'),
 ('6', 'd'),
 ('7', 'd'),
 ('8', 'd'),
 ('9', 'd'),
 ('10', 'd'),
 ('J', 'd'),
 ('Q', 'd'),
 ('K', 'd'),
 ('A', 'd')],
 [('2', 'c'),
 ('3', 'c'),
 ('4', 'c'),
 ('5', 'c'),
 ('6', 'c'),
 ('7', 'c'),
 ('8', 'c'),
 ('9', 'c'),
 ('10', 'c'),
 ('J', 'c'),
 ('Q', 'c'),
 ('K', 'c'),
 ('A', 'c')]]
```

```
In [285]: 1 print([list(zip(ranks, s * 13)) for s in suits])
```

```
[(('2', 's'), ('3', 's'), ('4', 's'), ('5', 's'), ('6', 's'), ('7', 's'), ('8', 's'), ('9', 's'), ('10', 's'), ('J', 's'), ('Q', 's'), ('K', 's'), ('A', 's'))], [('2', 'h'), ('3', 'h'), ('4', 'h'), ('5', 'h'), ('6', 'h'), ('7', 'h'), ('8', 'h'), ('9', 'h'), ('10', 'h'), ('J', 'h'), ('Q', 'h'), ('K', 'h'), ('A', 'h')], [('2', 'd'), ('3', 'd'), ('4', 'd'), ('5', 'd'), ('6', 'd'), ('7', 'd'), ('8', 'd'), ('9', 'd'), ('10', 'd'), ('J', 'd'), ('Q', 'd'), ('K', 'd'), ('A', 'd')], [('2', 'c'), ('3', 'c'), ('4', 'c'), ('5', 'c'), ('6', 'c'), ('7', 'c'), ('8', 'c'), ('9', 'c'), ('10', 'c'), ('J', 'c'), ('Q', 'c'), ('K', 'c'), ('A', 'c')]]
```

In [286]:

```
1 def f(list1, *, reverse = False):
2     if not reverse:
3         return sorted(list1)
4     else:
5         return sorted(list1, reverse = True)
6
7
8 print(f([3, 2, 4, 5, 1, 6, 8, 7]))
9 print(f([3, 2, 4, 5, 1, 6, 8, 7], reverse = True))
10 print(f([3, 2, 4, 5, 1, 6, 8, 7])[0])
11 print(f([3, 2, 4, 5, 1, 6, 8, 7], reverse = True)[0])
```

```
[1, 2, 3, 4, 5, 6, 7, 8]
[8, 7, 6, 5, 4, 3, 2, 1]
1
8
```

In [287]:

```
1 def f(list1, *, reverse = False):
2     return sorted(list1, reverse = reverse)
3
4
5 print(f([3, 2, 4, 5, 1, 6, 8, 7]))
6 print(f([3, 2, 4, 5, 1, 6, 8, 7], reverse = True))
7 print(f([3, 2, 4, 5, 1, 6, 8, 7])[0])
8 print(f([3, 2, 4, 5, 1, 6, 8, 7], reverse = True)[0])
```

```
[1, 2, 3, 4, 5, 6, 7, 8]
[8, 7, 6, 5, 4, 3, 2, 1]
1
8
```

In [288]:

```
1 print(hex(ord(chr(0xE3).upper())))
```

0xc3

In [289]:

```
1 print("abc" in "defabcghi")
```

True

In [290]:

```
1 print("abc" in "abbde")
```

False

In [291]:

```
1 def solve(a, b, c):
2     solution = ((-1 * b) + math.sqrt((b * b) - (4 * a * c))) / (2 * a)
3     return round(solution, 2)
4
5
```

```
6 print(solve(3, 4, -5))
7 print(solve(1, -5, -8))
```

```
0.79
6.27
```

```
In [292]: 1 def encrypt(s):
          2     return "".join(chr(ord(c) + 10) for c in s)
          3
          4
          5 print(encrypt("ABCDEFGHIJKLMNOPQRSTUVWXYZ"))
```

```
KL MN OP QR ST UV WX YZ [\]^_`abcd
```

```
In [293]: 1 def decrypt(s):
          2     return "".join(chr(ord(c) - 10) for c in s)
          3
          4
          5 print(decrypt("KL MN OP"))
          6 print(decrypt("S}kkm*%o\x81~yx"))
```

```
ABCDEF
Isaac Newton
```

```
In [294]: 1 print(10 % 3)
```

```
1
```

```
In [295]: 1 a = array.array("l", [20])
          2 print(a)
```

```
array('l', [20])
```

```
In [296]: 1 for i in a:
          2     print(i)
```

```
20
```

```
In [297]: 1 print(sys.getsizeof(a))
```

```
88
```

```
In [298]: 1 try:
          2     print(sys.getallocatedblocks(a))
          3 except TypeError as e:
          4     print(type(e).      )
```

```
TypeError
```

```
In [299]: 1 print(sys.getsizeof(1))
```

```
28
```

```
In [300]: 1 def f(s):
          2     print(s)
          3
```

```

4
5 try:
6     f(*"hello")
7 except TypeError as ex:
8     print(type(ex).      , ex.args)

```

TypeError ('f()' takes 1 positional argument but 5 were given',)

```

In [301]: 1 def func(s):
2           print(s)
3
4
5 def apply(func, *str):
6     result = func(*str)
7     return result
8
9
10 apply(func, "john")

```

john

```

In [302]: 1 try:
2           apply(func, "john", "smith")
3 except TypeError as ex:
4     print(type(ex).      , *ex.args)

```

TypeError func() takes 1 positional argument but 2 were given

```

In [303]: 1 def encloseGetPower(b):
2           print("hex(id(b)):", hex(id(b)))
3
4           def getPower(e):
5               print("hex(id(e)):", hex(id(e)))
6
7               return b ** e
8
9           print("getPower:", getPower)
10
11          return getPower
12
13
14 getBinaryPower = encloseGetPower(2)
15 print(getBinaryPower(31))
16 getTertiaryPower = encloseGetPower(3)
17 print(getTertiaryPower(4))

```

```

hex(id(b)): 0xad7e50
getPower: <function encloseGetPower.<locals>.getPower at 0x71b3f4643b60>
hex(id(e)): 0xad81f0
2147483648
hex(id(b)): 0xad7e70
getPower: <function encloseGetPower.<locals>.getPower at 0x71b3f4643c10>
hex(id(e)): 0xad7e90
81

```

```
In [304]: 1 print("getBinaryPower:", getBinaryPower)
          2 print("encloseGetPower:", encloseGetPower)
          3 try:
          4     print("getPower:", getPower)
          5 except NameError as ex:
          6     print(type(ex).      , ex.args)
```

```
getBinaryPower: <function encloseGetPower.<locals>.getPower at 0x71b3f4643b60>
encloseGetPower: <function encloseGetPower at 0x71b3f4643d70>
NameError ("name 'getPower' is not defined",)
```

```
In [305]: 1 print("getBinaryPower:", getBinaryPower.      )
          2 print("encloseGetPower:", encloseGetPower.      )
```

```
getBinaryPower: (<cell at 0x71b3f06e8c70: int object at 0xad7e50>,)
encloseGetPower: None
```

```
In [306]: 1 def outer(a, b):
          2     print(hex(id(a)))
          3     sumTotal = a + b
          4
          5     def inner():
          6         prod = a * b
          7         print(a, b, sumTotal, prod)
          8         return "You just called a closure!"
          9
         10     return inner
         11
         12
         13 f = outer(2, 3)
```

```
0xad7e50
```

```
In [307]: 1 f = outer(2, 3)
```

```
0xad7e50
```

```
In [308]: 1 print(f)
```

```
<function outer.<locals>.inner at 0x71b3f4642f00>
```

```
In [309]: 1 print(f.      )
```

```
(<cell at 0x71b3f06e88e0: int object at 0xad7e50>, <cell at 0x71b3f06e8730: int object at 0xad7e70>,
<cell at 0x71b3f06e8880: int object at 0xad7eb0>)
```

```
In [310]: 1 f()
```

```
2 3 5 6
```

```
Out[310]: 'You just called a closure!'
```

```
In [311]: 1 def execute(func):
          2
          3     def inner(a, b):
```

```
4         return func(a, b)
5
6     return inner
7
8
9 def add(a, b):
10     return a + b
11
12
13 add_executor = execute(add)
14 print(add_executor.)
```

(<cell at 0x71b3f46b0130: function object at 0x71b3f4642cf0>,)

In [312]: 1 print(hex(id(add)))

0x71b3f4642cf0

In [313]: 1 try:
2 print(hex(add))
3 except TypeError as ex:
4 print(type(ex).)

TypeError

In [314]: 1 add_executor(2, 3)

Out[314]: 5

In [315]: 1 def f():
2 pass
3
4
5 print(f.)

None

In [316]: 1 def f():
2
3 def g():
4 return
5
6 return g
7
8
9 h = f()
10 print(f.)
11 print(h.)

None

None

In [317]: 1 a = 0
2 try:

```
3     print(a.          )
4 except AttributeError as e:
5     print(type(e).    )
```

AttributeError

```
In [318]: 1 a = []
2 try:
3     print(a.          )
4 except AttributeError as e:
5     print(type(e).    )
```

AttributeError

```
In [319]: 1 def execute(func):
2
3         def inner(*posargs, **kwargs):
4             return func(*posargs, **kwargs)
5
6         return inner
7
8
9 def add(a, b, c):
10     return a + b + c
11
12
13 def say_hello(name, *, formal = True):
14     return f"Pleased to meet you, {name}" if formal else f"Hi, {name}!"
15
16
17 exec_add = execute(add)
18 exec_greet = execute(say_hello)
```

```
In [320]: 1 exec_add(1, 2, 3)
```

Out[320]: 6

```
In [321]: 1 exec_greet("Michael", formal = False)
```

Out[321]: 'Hi, Michael!'

```
In [322]: 1 exec_greet("Michael", formal = True)
```

Out[322]: 'Pleased to meet you, Michael'

```
In [323]: 1 print((0.1).as_integer_ratio())
```

(3602879701896397, 36028797018963968)

```
In [324]: 1 print((0.125).as_integer_ratio())
```

(1, 8)

```
In [325]: 1 print((0.2).as_integer_ratio())
          2 print(f"{0.2:.40f}")
```

```
(3602879701896397, 18014398509481984)
0.20000000000000000111022302462515654042363
```

```
In [326]: 1 print(math.pi.as_integer_ratio())
          2 n, d = math.pi.as_integer_ratio()
          3 print(n / d)
```

```
(884279719003555, 281474976710656)
3.141592653589793
```

```
In [327]: 1 try:
          2     a1 = a1 + 1
          3     print(a1)
          4 except NameError as ex:
          5     print(type(ex).      , ": ", *ex.args, ".", sep = "")
```

```
NameError: name 'a1' is not defined.
```

```
In [328]: 1 def getDiagonalMatrix(val, rows, columns):
          2     diagonalMatrix = [[val if ce == r else 0 for ce in range(4)] for r in
          3     range(5)]
          4     return diagonalMatrix
```

```
In [329]: 1 def getDiagonalMatrix(rows, columns, *, val):
          2     diagonalMatrix = [[val if cl == row else 0 for cl in range(columns)]
          3     for row in range(rows)]
          4     return diagonalMatrix
```

```
In [330]: 1 def getFactorial(n):
          2     factorial = 1
          3     for i in range(1, n + 1):
          4         factorial *= i
          5     return factorial
```

```
In [331]: 1 result = getFactorial(5)
          2 print(result)
```

```
120
```

```
In [332]: 1 result = getDiagonalMatrix(3, 4, val = 5)
          2 pprint.pprint(result, width = 14)
```

```
[[5, 0, 0, 0],
 [0, 5, 0, 0],
 [0, 0, 5, 0]]
```

```
In [333]: 1 print(f"{timeit.timeit('getFactorial(2)', globals = globals()):.4g}")
```

```
0.1599
```

```
In [334]: 1 n = 0.000000000000123456789
2 n = 4.482882e-05
3 exp = 0
4 while n < 1:
5     exp -= 3
6     n *= 1000
7 prefixes = "", "m", "\N{GREEK SMALL LETTER MU}", "n", "p", "f", "a"
8 print(f"{n}{prefixes[abs(exp)//3]}")
```

44.82882μ

```
In [335]: 1 n = 1000000000
2 exp = 0
3 while n >= 1000:
4     exp += 3
5     n /= 1000
6 prefixes = "", "k", "M", "G", "T", "P", "E"
7 print(f"{n}{prefixes[exp//3]}")
```

1.0G

```
In [336]: def toEnggNotation(n, scientificNotation = True, nSignificantDigits = 4):
1     exp = 0
2     oldN = n
3     if n > 1:
4         while n >= 1000:
5             exp += 3
6             n /= 1000
7         prefixes = "", "k", "M", "G", "T", "P", "E"
8         return f"{n:.3f}{prefixes[exp//3]}\t{oldN}" if
9 scientificNotation else f"{n}{prefixes[exp//3]}"
10    elif n < 1:
11        while n < 1:
12            exp -= 3
13            n *= 1000
14        prefixes = "", "m", "\N{GREEK SMALL LETTER MU}", "n", "p", "f",
15 "a"
16        return f"{n:.{nSignificantDigits}f}
17 {prefixes[abs(exp)//3]}\t{oldN}" if scientificNotation else f"{n}
18 {prefixes[abs(exp)//3]}"
19    else:
20        return 1
21
22
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96
97
98
99
```

```
1 def myTimer(f, *pos, **keyword):
2     start = time.perf_counter()
3     print(f(*pos, **keyword))
4     end = time.perf_counter()
5     return end - start
6
7
```

```
8 print(toEnggNotation(myTimer(getFactorial, 1000)))
9 print(toEnggNotation(myTimer(getDiagonalMatrix, 3, 5, val = 7)))
```

[illegible]

```

1 def encloseMyTimer(f):
2
3     def myTimer(*pos, **keyword):
4         start = time.perf_counter()
5         result = f(*pos, **keyword)
6         end = time.perf_counter()
7         print(toEnggNotation(end - start, True, 4))
8         return result
9
10    return myTimer
11
12
13 timed = encloseMyTimer(getDiagonalMatrix)
14 print(timed(2, 4, val = -5))
15 timed = encloseMyTimer(getFactorial)
16 print(timed(11))
17 print(encloseMyTimer(getDiagonalMatrix)(2, 4, val = -5))
18 print(encloseMyTimer(getFactorial)(11))

```

```
4.1970µ 4.19699790654704e-06
[[-5, 0, 0, 0], [0, -5, 0, 0]]
2.2940µ 2.2940002963878214e-06
39916800
2.3040µ 2.3040047381073236e-06
[[-5, 0, 0, 0], [0, -5, 0, 0]]
1.4520µ 1.4519973774440587e-06
39916800
```

```
In [341]: 1 # print(eval("2 * 3"))
          2 # print(eval("x = 2 and print(x)"))
```

```
In [342]: 1 # eval("x = 2")
          2 # print(x)
```

```
In [343]: 1 x = 2
          2 eval("x ** 2 - 1")
```

```
Out[343]: 3
```

```
In [344]: 1 def f(expr, x):
          2     print(eval(expr))
          3
          4
          5 x = 2
          6 f("x ** 2 - 1", x)
```

```
3
```

```
In [345]: 1 def f(x):
          2     return True
          3
          4
          5 try:
          6     print(list(map(f(False), range(0, 5))))
          7 except TypeError as ex:
          8     print(type(ex).      + ":", ex.args[0])
```

```
TypeError: 'bool' object is not callable
```

```
In [346]: 1 def getExtremum(expr, interval, granularity, minimax):
          2
          3     def f(expr):
          4
          5         def g(x):
          6             return eval(expr)
          7
          8         return g
          9
          10     evalExpr = f(expr)
          11     print("Range:", list(range(interval[0], interval[1] + granularity,
          12 granularity)))
          13     if minimax == "minimum":
```

```

7         return min(map(evalExpr, range(interval[0], interval[1] +
granularity, granularity)))
8         elif minimax == "maximum":
9             return max(map(evalExpr, range(interval[0], interval[1] +
granularity, granularity)))
1        else:
10            return "Invalid minimax"
11
12
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99

```

```

Range: [-10, -8, -6, -4, -2, 0, 2, 4, 6, 8, 10]
-1
Range: [-10, -8, -6, -4, -2, 0, 2, 4, 6, 8, 10]
99

```

```

In [347]: 1 expr = "x ** 2 - 1"
2         f = lambda x: eval(expr)
3         print(f(3))

```

8

```

In [348]: 1 def getExtremum(expr, interval, resolution, minimax):
2
3         def f(expr):
4             def g(x):
5                 return eval(expr)
6
7             return g
8
9         evalExpr = f(expr)

```

```

    l = [interval[0] + (i * (interval[1] - interval[0]) / (resolution -
6    1)) for i in range(resolution)] # Cheated with the calc.
        return min(map(evalExpr, l)) if minimax == "minimum" else
7    max(map(evalExpr, l))

8
    expr = "x ** 2 - 1"
9    print(getExtremum(expr, (-10, 10), 5, "minimum"))
1    print(getExtremum(expr, (-10, 10), 5, "maximum"))
0    print(getExtremum(expr, (-10, 10), 6, "minimum"))
1    print(getExtremum(expr, (-10, 10), 6, "maximum"))
1    expr = "math.log10(x)"
1    print(getExtremum(expr, (1, 100), 20, "minimum"))
2    print(getExtremum(expr, (1, 100), 20, "maximum"))
1    expr = "math.sin(x)"
3    print(getExtremum(expr, (-10, 10), 20, "minimum"))
1    print(getExtremum(expr, (-10, 10), 20000, "maximum"))
4    expr = "x"
1    print(getExtremum(expr, (1, 6), 8, "minimum"))
5    print(getExtremum(expr, (1, 6), 8, "maximum"))

1
6
1
7
1
8
1
9
2
0
2
1
2
2
2
3
2
4
2
5
2
6
2
7
2
8

```

```
-1.0
99.0
3.0
99.0
0.0
2.0
-0.9999667804441291
0.999999996044897
1.0
6.0
```

In [349]:

```
def f(t):
    return t[0] ** t[1]

l = [(1, 1), (1, 2), (2, 1), (2, 2), (2, 3), (3, 2), (3, 3), (3, 4), (4, 3),
      (4, 4)]

start = time.perf_counter()
for _ in range(10_000):
    result = []
    for t in l:
        result.append(f(t))
end = time.perf_counter()
forLoopTime = end - start
print(result)

start = time.perf_counter()
for _ in range(10_000):
    result = [f(t) for t in l]
end = time.perf_counter()
listCompTime = end - start
print(result)

start = time.perf_counter()
for _ in range(10_000):
    result = list(map(f, l))
end = time.perf_counter()
mapTime = end - start
print(result)

print(toEnggNotation(forLoopTime), toEnggNotation(listCompTime),
      toEnggNotation(mapTime))
```

2
2
2
3
2
4
2
5
2
6
2
7
2
8
2
9
3
0

```
[1, 1, 2, 4, 8, 9, 27, 81, 64, 256]
[1, 1, 2, 4, 8, 9, 27, 81, 64, 256]
[1, 1, 2, 4, 8, 9, 27, 81, 64, 256]
12.7960m      0.012796039998647757 9.0565m      0.00905650600179797 9.7188m      0.00971875200048089
```

In [350]:

```
def power(x, y):
    return x ** y

def dist(pt1, pt2):
    return math.sqrt(sum(coord1 - coord2 for coord1, coord2 in zip(pt1,
pt2)))

def partial(f, operand2):

    def calcPartialFunction(operand1):
        return f(operand1, operand2)

    return calcPartialFunction

squares = partial(power, 2)
print(squares(3))
print(squares(4))
dist_from_origin = partial(dist, (0, 0))
print(dist_from_origin((1, 1)))
gcd_60 = partial(math.gcd, 60)
print(gcd_60(13))
log_2 = partial(math.log, 2)
print(log_2(8))
log_10 = partial(math.log, 10)
print(log_10(100))
```

```
5 log_16 = partial(math.log, 16)
6 print(log_16(65536))
```

```
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```

```
9
16
1.4142135623730951
1
3.0
2.0
4.0
```

In [351]:

```
1 def norm(x, y):
2     return math.sqrt(x ** 2 + y ** 2)
3
4
5 def find_index_min(seq):
6     minima = min(seq)
7     return seq.index(minima)
8
9
10 def logged(f):
11
12     def g(*args):
13         print("Logging...")
14         start = time.perf_counter()
15         print(f(*args))
```

```

16         end = time.perf_counter()
17         print(toEnggNotation(end - start))
18         print("... Finished")
19
20     return g
21
22
23 loggedNorm = logged(norm)
24 loggedNorm(3, 4)
25 loggedFindIndexMin = logged(find_index_min)
26 loggedFindIndexMin([8, 7, 2, 9, 4])

```

```

Logging...
5.0
19.1260µ      1.9126004190184176e-05
... Finished
Logging...
2
18.5750µ      1.857500319601968e-05
... Finished

```

```

In [352]: 1 def encloseGetPower(f):
2
3           def getPower(*args):
4               return f(*args)
5
6           print(hex(id(getPower)))
7           return getPower
8
9
10  getPower = encloseGetPower(pow)
11  print(getPower(2, 5))
12  print(hex(id(getPower)))
13  print(hex(id(encloseGetPower)))
14  print(getPower.          )
15  print(encloseGetPower.   )

```

```

0x71b3f06f9380
32
0x71b3f06f9380
0x71b3f06f8d50
(<cell at 0x71b438c1cd30: builtin_function_or_method object at 0x71b44ddb0630>,)
None

```

```

In [353]: 1 l = [(1, 2), (-4, -5.5), (10, -10), (-2, 2)]
2          print(sorted(l, key = lambda t: abs(t[0] - t[1]), reverse = True))

```

```

[(10, -10), (-2, 2), (-4, -5.5), (1, 2)]

```

```

In [354]: 1 l = ["s", "0", "1"]
2          card = "".join(reversed(l))
3          print(card)
4          card = ""
5          for k in reversed(l):

```

```
6     card += k
7     print(card)
```

```
10s
10s
```

In [355]:

```
1
2
3
4
5 suits = 'shdc' # Spades, Hearts, Diamonds, Clubs
6 ranks = list('23456789') + ['10', 'J', 'Q', 'K', 'A']
7 cards = [[rank + suit for rank in ranks] for suit in suits]
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
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96
97
98
99
```

```
[['As', 'Ks', 'Qs', 'Js', '10s', '9s', '8s', '7s', '6s', '5s', '4s', '3s', '2s'], ['Ah', 'Kh', 'Qh', 'Jh', '10h', '9h', '8h', '7h', '6h', '5h', '4h', '3h', '2h'], ['Ad', 'Kd', 'Qd', 'Jd', '10d', '9d', '8d', '7d', '6d', '5d', '4d', '3d', '2d'], ['Ac', 'Kc', 'Qc', 'Jc', '10c', '9c', '8c', '7c', '6c', '5c', '4c', '3c', '2c']]
[['2s', '3s', '4s', '5s', '6s', '7s', '8s', '9s', '10s', 'Js', 'Qs', 'Ks', 'As'], ['2h', '3h', '4h', '5h', '6h', '7h', '8h', '9h', '10h', 'Jh', 'Qh', 'Kh', 'Ah'], ['2d', '3d', '4d', '5d', '6d', '7d', '8d', '9d', '10d', 'Jd', 'Qd', 'Kd', 'Ad'], ['2c', '3c', '4c', '5c', '6c', '7c', '8c', '9c', '10c', 'Jc', 'Qc', 'Kc', 'Ac']]
```

In [356]:

```
1 data = {'S1': (100, 200, 80, 180), 'S2': (10, 20, 8, 18), 'S3': (50, 150, 50, 150)}
```

```

2
3 def findMinima(data, reverse = False):
4     minimax = min if reverse else max
5     return minimax(data, key = lambda d: data[d][1] - data[d][2])
6
7
8
9
10 print(findMinima(data, reverse = True))
11 print(findMinima(data))
12

```

S2
S1

In [357]:

```

1 quotes = [('AACC', 6.05, 6.07, 6.03, 6.05, 65800), ('AAME', 1.7, 1.82, 1.7,
2 1.82, 4300), ('AAON', 24.98, 25.07, 24.9, 24.94, 28200), ('AAPL', 317.99,
3 319.57, 316.75, 317.13, 12901800), ('AATI', 3.82, 3.82, 3.74, 3.79, 194600),
4 ('AAWW', 60.89, 61.44, 60.5, 61.19, 272800), ('AAXJ', 65.4, 65.71, 65.28,
5 65.56, 390300), ('ABAT', 4.01, 4.01, 3.95, 3.99, 656300), ('ABAX', 25.26,
6 25.49, 25.04, 25.42, 73700), ('ABBC', 11.75, 11.88, 11.48, 11.53, 29700),
7 ('ABCB', 9.3, 9.3, 9.06, 9.14, 42600), ('ABCD', 3.25, 3.25, 3.11, 3.22,
8 122800), ('ABCO', 48.75, 50.41, 46.9, 50.37, 66300), ('ABCW', 0.52, 0.61, 0.52,
9 0.53, 83000), ('ABFS', 25.98, 26.27, 25.41, 25.5, 384900), ('ABIO', 3.96, 4,
10 3.88, 4, 38500), ('ABMD', 11.94, 12, 11.69, 11.87, 122600), ('ABTL', 0.82,
1 0.84, 0.82, 0.83, 28700), ('ABVA', 3.09, 3.25, 3.09, 3.25, 6200), ('ACAD',
2 0.76, 0.76, 0.7, 0.74, 341500), ('ACAS', 7.52, 7.72, 7.52, 7.66, 5199800),
3 ('ACAT', 14.44, 14.44, 14.04, 14.2, 51700), ('ACCL', 8.11, 8.21, 7.94, 8.1,
4 456100), ('ACET', 8.01, 8.04, 7.13, 7.73, 575600), ('ACFC', 1.69, 1.7, 1.5,
5 1.6, 12300), ('ACFN', 3.82, 4, 3.82, 3.98, 53700), ('ACGL', 89.76, 90.14,
6 89.39, 89.92, 240900), ('ACGY', 22.41, 22.56, 22.25, 22.46, 86800), ('ACHN',
7 3.12, 3.2, 3.07, 3.16, 113700), ('ACIW', 26.96, 27.03, 26.63, 26.8, 157000),
8 ('ACLI', 33.65, 33.77, 33.45, 33.63, 28700), ('ACLS', 2.47, 2.63, 2.46, 2.53,
9 1818800), ('ACMR', 2.69, 2.84, 2.37, 2.71, 158600), ('ACOM', 25.2, 26.6, 24.9,
10 26.56, 265300), ('ACOR', 26.67, 27.07, 26.38, 27.04, 1415000), ('ACPW', 1.84,
1 1.89, 1.77, 1.85, 565500), ('ACTG', 27.2, 27.43, 26.86, 27.18, 228800),
2 ('ACTI', 3.25, 3.26, 3.25, 3.26, 148500), ('ACTS', 2.08, 2.09, 2.07, 2.07,
3 130500), ('ACUR', 2.6, 2.64, 2.51, 2.6, 16000), ('ACWI', 46.53, 46.7, 46.32,
4 46.51, 286200), ('ACWX', 44.49, 44.66, 44.36, 44.6, 55500), ('ACXM', 18, 18.07,
5 17.81, 18.01, 289800), ('ADAM', 7.34, 7.49, 7.33, 7.44, 81700), ('ADAT', 0.6,
6 0.68, 0.59, 0.66, 86400), ('ADBE', 29.43, 29.71, 29.07, 29.14, 7585300),
7 ('ADCT', 12.68, 12.69, 12.66, 12.68, 1660500), ('ADEP', 6.14, 6.14, 4.95, 5.61,
8 71000), ('ADES', 6.2, 6.22, 6, 6.19, 4800), ('ADGF', 4.31, 4.55, 4.31, 4.54,
9 10200)]

```

```
def myFilter(percentage = 10):
    return list(filter(lambda t: abs((t[4] - t[2]) / t[2]) > (percentage / 100), quotes))

print("10:", myFilter(10))
print("5:", myFilter(5))
print("1:", myFilter(1))
```

```
10: [('ABCW', 0.52, 0.61, 0.52, 0.53, 83000)]
5: [('ABCW', 0.52, 0.61, 0.52, 0.53, 83000), ('ACFC', 1.69, 1.7, 1.5, 1.6, 12300), ('ADEP', 6.14, 6.14, 4.95, 5.61, 71000)]
1: [('ABBC', 11.75, 11.88, 11.48, 11.53, 29700), ('ABCB', 9.3, 9.3, 9.06, 9.14, 42600), ('ABCW', 0.52, 0.61, 0.52, 0.53, 83000), ('ABFS', 25.98, 26.27, 25.41, 25.5, 384900), ('ABMD', 11.94, 12, 11.69, 11.87, 122600), ('ABTL', 0.82, 0.84, 0.82, 0.83, 28700), ('ACAD', 0.76, 0.76, 0.7, 0.74, 341500), ('ACAT', 14.44, 14.44, 14.04, 14.2, 51700), ('ACCL', 8.11, 8.21, 7.94, 8.1, 456100), ('ACET', 8.01, 8.04, 7.13, 7.73, 575600), ('ACFC', 1.69, 1.7, 1.5, 1.6, 12300), ('ACHN', 3.12, 3.2, 3.07, 3.16, 113700), ('ACLS', 2.47, 2.63, 2.46, 2.53, 1818800), ('ACMR', 2.69, 2.84, 2.37, 2.71, 158600), ('ACPW', 1.84, 1.89, 1.77, 1.85, 565500), ('ACUR', 2.6, 2.64, 2.51, 2.6, 16000), ('ADAT', 0.6, 0.68, 0.59, 0.66, 86400), ('ADBE', 29.43, 29.71, 29.07, 29.14, 7585300), ('ADEP', 6.14, 6.14, 4.95, 5.61, 71000)]
```

```
In [358]: 1 l = [5, 6, -4, 8]
          2 print(min(l, key = abs))
```

```
-4
```

```
In [359]: 1 importlib.reload(logging)
          2 logging.basicConfig(format = "%(asctime)s %(levelname)s %(message)s", datefmt = "%FT%Tb")
          3 logging.error("This is an error")
```

```
2026-09-08T18:37:38b ERROR This is an error
```

```
In [360]: 1 logger = logging.getLogger( )
          2 logger.setLevel(logging.DEBUG)
          3 logger.error("This is an error message")
          4 logger.debug("This is a debug message")
```

```
2026-09-08T18:37:38b ERROR This is an error message
2026-09-08T18:37:38b DEBUG This is a debug message
2026-09-08T18:37:38b DEBUG This is a debug message
```

```
In [361]: 1 def f():
          2     pass
          3
          4
          5 print(f. )
          6 print(type(f. ))
```

```
f
<class 'str'>
```

```
In [362]: 1 def partial(f):
          2
          3     def callF(*pargs, **kwargs):
```

```

4         logger.debug("Calling your function... " + f.           )
5         start = time.perf_counter()
6         result = f(*pargs, **kwargs)
7         end = time.perf_counter()
8         logger.debug("... Function returned.")
9         logger.info(f"Elapsed Time: {toEnggNotation(end - start)}")
10        # print(f"{toEnggNotation(end - start)}")
11        return result
12
13    return callF
14
15
16    def add(a, b):
17        return a + b
18
19
20    def log3(a):
21        return math.log(a, 3)
22
23
24    @partial
25    def log4(a):
26        return math.log(a, 4)
27
28
29    myAdder = partial(add)
30    logger.info("myAdder: " + str(myAdder(2, 3)))
31    log3 = partial(log3)
32    logger.info("log3: " + str(log3(9)))
33    logger.info("log4: " + str(log4(16)))

```

```

2026-09-08T18:37:38b DEBUG Calling your function... add
2026-09-08T18:37:38b DEBUG ... Function returned.
2026-09-08T18:37:38b INFO Elapsed Time: 1.9940µ 1.9939980120398104e-06
2026-09-08T18:37:38b INFO myAdder: 5
2026-09-08T18:37:38b DEBUG Calling your function... log3
2026-09-08T18:37:38b DEBUG ... Function returned.
2026-09-08T18:37:38b DEBUG ... Function returned.
2026-09-08T18:37:38b INFO Elapsed Time: 1.9940µ 1.9939980120398104e-06
2026-09-08T18:37:38b INFO myAdder: 5
2026-09-08T18:37:38b DEBUG Calling your function... log3
2026-09-08T18:37:38b DEBUG ... Function returned.
2026-09-08T18:37:38b INFO Elapsed Time: 5.8610µ 5.861002136953175e-06
2026-09-08T18:37:38b INFO log3: 2.0
2026-09-08T18:37:38b DEBUG Calling your function... log4
2026-09-08T18:37:38b DEBUG ... Function returned.
2026-09-08T18:37:38b INFO Elapsed Time: 4.8090µ 4.80900052934885e-06
2026-09-08T18:37:38b INFO log4: 2.0

```

In [363]:

```

1    def cache(f):
2        print("Initializing cache...")
3        cacheDict = {}
4
5        def callF(*pargs):

```

```

6         if pargs in cacheDict:
7             print("Cache Hit")
8             return cacheDict[pargs]
9         else:
10            print("Cache Miss")
11            result = f(*pargs)
12            cacheDict[pargs] = result
13            return result
14
15    return callF
16
17
18    @cache
19    def add(a, b):
20        return a + b
21
22
23    @cache
24    def mul(a, b):
25        return a * b

```

Initializing cache...
Initializing cache...

```

In [364]: 1 print(add.      )
          2 print(mul.     )

```

(<cell at 0x71b3f06e9300: dict object at 0x71b3f0737400>, <cell at 0x71b3f06ea1a0: function object at 0x71b3f0757ab0>)
(<cell at 0x71b3f06ead10: dict object at 0x71b3f0737440>, <cell at 0x71b3f06e8fa0: function object at 0x71b3f0757cc0>)

```

In [365]: 1 print(add(2, 3))
          2 print(mul(2, 3))
          3 print(add(2, 3))
          4 print(mul(2, 3))

```

Cache Miss
5
Cache Miss
6
Cache Hit
5
Cache Hit
6

```

In [366]: 1 print(type(func tools.lru_cache))

```

<class 'function'>

```

In [367]: 1 @func tools.lru_cache(maxsize = 2)
          2 def graham(a, b, c):
          3     print("Cache Miss")
          4     return a ** b ** c
          5

```

```
6  
7 print(gramham(2, 2, 4))
```

Cache Miss
65536

```
In [368]: 1 print(gramham(2, 2, 4))  
          2 print(gramham(2, 2, 4))  
          3 print(gramham(2, 2, 5))  
          4 print(gramham(2, 2, 6))  
          5 print(gramham(2, 2, 4))
```

65536
65536
Cache Miss
4294967296
Cache Miss
18446744073709551616
Cache Miss
65536

```
In [369]: 1 @functools.lru_cache(maxsize = 100)  
          2 def fib(n):  
          3     print(f"Calling fib{n}...")  
          4     if n <= 1:  
          5         return n  
          6     return fib(n - 1) + fib(n - 2)  
          7  
          8  
          9 print(fib(2000))
```

Calling fib2000...
Calling fib1999...
Calling fib1998...
Calling fib1997...
Calling fib1996...
Calling fib1995...
Calling fib1994...
Calling fib1993...
Calling fib1992...
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2295243347964480139515349562972087652656069529806499841977448720155612802665404554171717881930324025
204312082516817125
```

In [370]:

```
1 def myTimer(f):
2
3     def callF(*args):
4         start = time.perf_counter()
5         result = f(*args)
```

```

6         end = time.perf_counter()
7         print(toEnggNotation(end - start))
8         return result
9
10        return callF
11
12
13    # timedPower = myTimer(math.pow)
14    # print(timedPower(5, 188))
15    #
16    #
17    @myTimer
18    def myPower(a, b):
19        return a ** b
20
21
22    print(myPower(5, 188))

```

```

2.8350µ 2.834996848832816e-06
2548947057811923643246208636428338888945767288308190057718217365417860642420893057508943799552292297
50557351508177816867828369140625

```

```
In [20]: 1 PRECISION = 4
```

```
In [21]: 1 a = decimal.Decimal(math.e)
2 print(a)
3 print(type(a))
4 print(float(a))
```

```

2.718281828459045090795598298427648842334747314453125
<class 'decimal.Decimal'>
2.718281828459045

```

```
In [22]: 1 a = decimal.Decimal("2.404040313893824388847472828")
2 print(a)
3 print(type(a))
4 print(float(a))
```

```

2.404040313893824388847472828
<class 'decimal.Decimal'>
2.4040403138938244

```

```
In [23]: 1 a = decimal.Decimal(2.2384892948592929275758693)
2 print(a)
3 print(type(a))
4 print(float(a))
```

```

2.23848929485929293292656439007259905338287353515625
<class 'decimal.Decimal'>
2.238489294859293

```

```
In [24]: 1 l = [decimal.Decimal(random.random()) for _ in range(10)]
2 a = sum(l) / len(l)
```

```
3 print(a)
4 print(type(a))
```

```
0.5607448911234186317287253586
<class 'decimal.Decimal'>
```

```
In [ ]: 1 def normalize(f):
2
3         def callF(*args):
4             result = f(*args)
5             # print(type(result))
6             result = float(result)
7             return float(f"{result:.{PRECISION}f}")
8
9         return callF
```

```
In [26]: 1 @normalize
2 def perc_diff(x, y):
3     try:
4         return (y - x) / x * 100
5     except ZeroDivisionError:
6         return 0
7
8
9 @normalize
10 def sum_squares(*args):
11     return sum(x ** 2 for x in args)
12
13
14 @normalize
15 def avg(*args):
16     if len(args) == 0:
17         return 0
18
19     numbers = [decimal.Decimal(x) for x in args]
20     sum_ = sum(numbers)
21     return sum_ / len(numbers)
```

```
In [27]: 1 perc_diff(5088, 2872)
```

```
<class 'float'>
```

```
Out[27]: -43.5535
```

```
In [28]: 1 avg(2.4, 4.1, 5.932929, 8.2838485)
```

```
<class 'decimal.Decimal'>
```

```
Out[28]: 5.1792
```

```
In [29]: 1 def add(x, y):
2         time.sleep(2)
3         return x + y
4
```

```
5  
6 print(timeit.timeit("add(2340320832580385043843805805,  
320342408320482804320584305)", globals = globals(), number = 10))
```

20.000951050998992

In [30]:

```
1 @functools.lru_cache  
2 def add(x, y):  
3     time.sleep(2)  
4     return x + y  
5  
6  
7 print(timeit.timeit("add(2340320832580385043843805805,  
320342408320482804320584305)", globals = globals(), number = 10))
```

2.000115867005661